March/April 2015

InTech

A PUBLICATION OF THE INTERNATIONAL SOCIETY OF AUTOMATION

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Industrial automation systems cybersecurity

Embedding end-to-end trust and security

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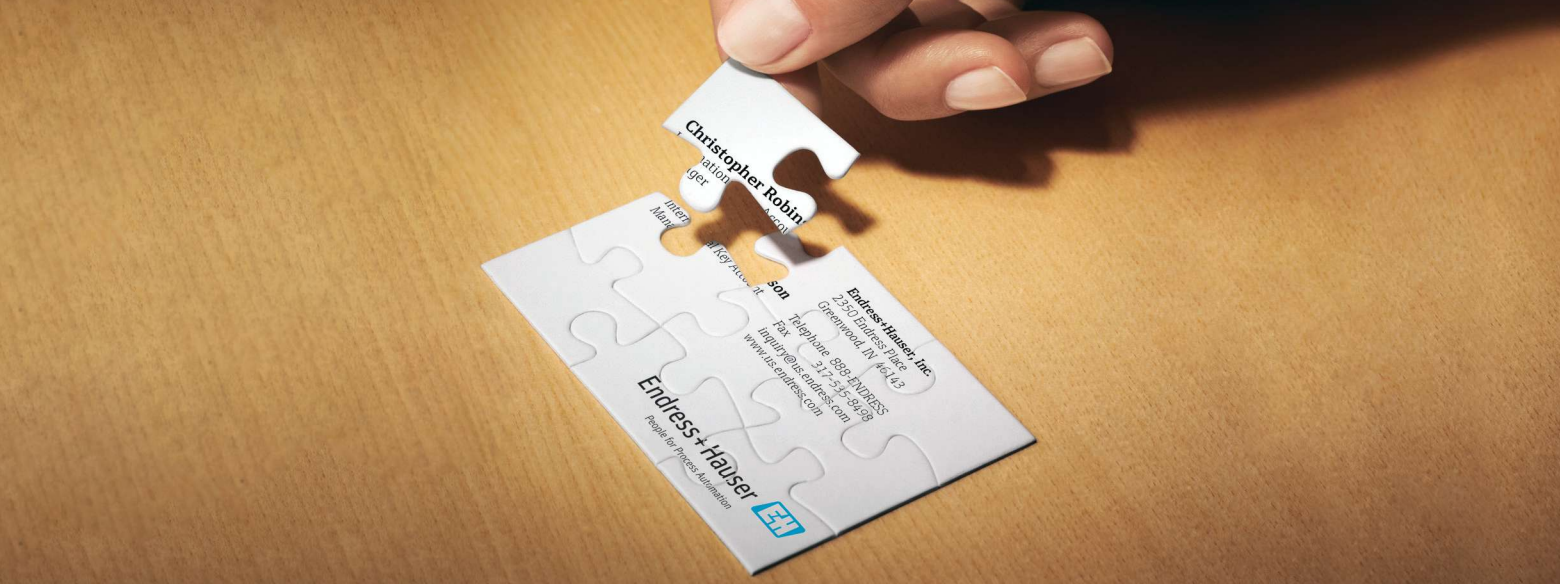

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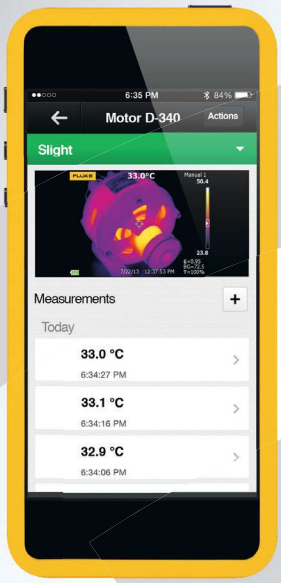

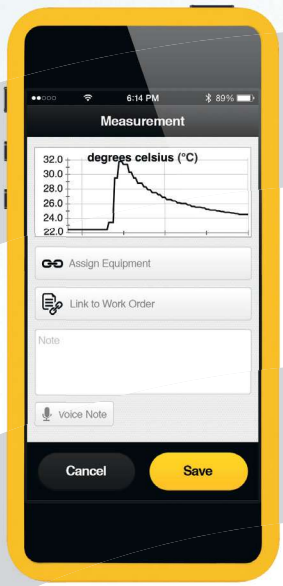

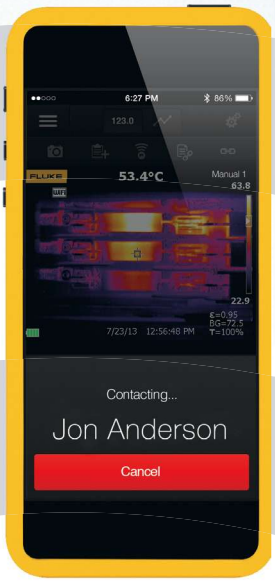

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By Sven Schrecker

Today's data-driven, interconnected world has changed the security landscape for industrial automation and control systems. This article discusses the current state of industrial automation system security, the technological and organizational challenges to improving it, and a dynamic model for embedding end-to-end trust and security into ICS and SCADA systems.

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WEB EXCLUSIVE

Industrial wireless sensor networks

The industrial wireless sensor networking value proposition is evolving from extending or replacing wired networks to cloud-connected smart-object intelligence. Wireless sensor adoption is accelerating due to a new generation of standards-based industrial networks, according to ON World's latest survey.

Read more at:
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InTech provides the most thought-provoking and authoritative coverage of automation technologies, applications, and strategies to enhance automation professionals' on-the-job success. Published by the industry's leading organization, ISA, *InTech* addresses the most critical issues facing the rapidly changing automation industry.

New possibilities with Beamex workshop

The Beamex MCS200 Workshop offers a controlled environment enabling very accurate calibration and ergonomic ready-to-use equipment. Beamex has recently launched the product, MC6 Workstation, an advanced panel mounted documenting calibrator and communicator. MC6 Workstation offers calibration capabilities for pressure, temperature and various electrical signals. Full fieldbus communication for HART, FOUNDATION Fieldbus and Profibus PA instruments is available. MC6 Workstation communicates with pressure controllers and temperature blocks, enabling fully automatic calibration.

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Be a voice setting industry direction

By Bill Lydon, *InTech*, Chief Editor

In my discussions with users, they express the desire for automation and control systems with useful, consistent features and interfaces that satisfy real-world needs. Automation professionals have the opportunity to influence and improve automation industry directions when they are active members of ISA, participating in the development of high-quality, unbiased automation information, standards, and technical information.

A great example is the ISA-88 series of standards, the first published 20 years ago, that has improved productivity, efficiency, and quality for manufacturers worldwide and inspired other standards, including the ISA-95 series. Before ISA-88, each automation manufacturer had a unique method and system to organize and execute batch process automation. This required users to learn the application philosophy and specifics of systems from each automation vendor. The ISA-88 standards for Batch Control include a common set of terminology, definitions, and data structures, as well as a definition of the physical, process, and procedural models of a batch manufacturing system. These standard guidelines are general in nature and were developed specifically for batch industries, such as specialty chemicals, pharmaceuticals, food and beverage, and consumer product industries. Applying these standards properly can allow for highly flexible manufacturing that is easily configured, maintained, and modified.

Due to the success of the widely accepted standards, the concept has been applied to continuous processes and discrete manufacturing. Acknowledged benefits of ISA-88 adoption in the real world include quicker time to market, faster new plant startups, reliable product changeovers, the ability to create recipes in parallel, lower total cost of automation ownership, and reduced training costs as engineers move between plants and process cells with different suppliers' systems and processes. The standards have also empowered chemists and food scientists and others to develop recipes and make processing changes without having to understand detailed programming and equipment functions.

Your opportunity

Automation industry improvements such as ISA-88 come from the active participation of automation professionals in developing standards and best practices that satisfy real industry needs. Your opportunity is to participate in several key ISA initiatives to improve the automation industry.

Your contributions to the topics of the day are valuable. They include setting engineering and technology standards to improve the productivity, efficiency, management, safety, and cybersecurity of modern automation and control systems. Standards help clarify user needs for suppliers, providing structured input of requirements, which influence suppliers to develop the most effective industrial automation products.

The next time you are complaining about the difficulty of using industrial automation and control products, think about your opportunity to improve the industrial automation industry and make things better. You have a voice, and this is a valuable experience for yourself that you should seriously consider. ISA gives members a way to work together to develop and deliver high-quality, unbiased automation information, standards, and technical information and opportunities to network with other automation professionals.

More than 4,000 individuals working across 140 committees and subgroups from cybersecurity to process safety to interfaces between industrial process control computers and subsystems are developing standards in key areas. Volunteers are not asked to make ISA their main career goal, but everyone's contribution is valuable. ■

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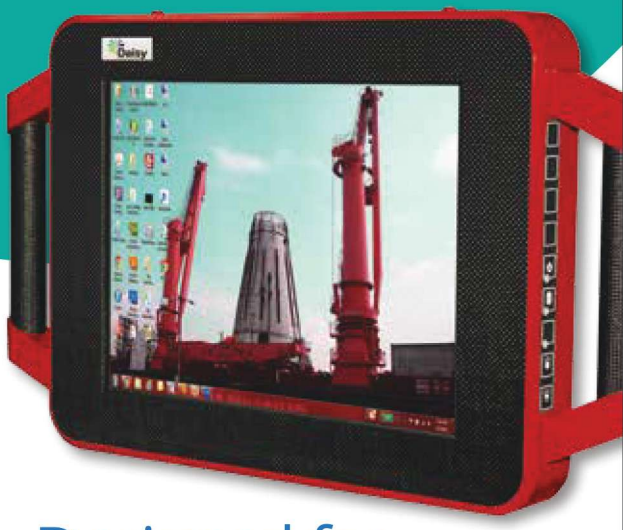

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Editor's note: In the January/February 2015 issue of *InTech*, we published a letter that challenged Peter Martin's comments about generational differences in a Final Say column (September/October 2014 "The right generation at the right time"). Here Martin responds to the letter.

Generational differences

I tend to agree with a number of the points Mr. Drobny made in reference to my article in *InTech*. I do believe there has been a sliding shift in generational perspectives that is age related and not specific-era related. I also agree that using terms such as "baby boomers" and "millennials" may be feeding into an era-level perspective and can be damaging if misused. I suppose I should point out that I did not invent these terms, rather I used them to try to make similar points to those Mr. Drobny makes.

I have heard much banter across industry, primarily by more experienced industrial people, targeting the younger generation (millennials) as a huge problem for industrial operations. They point out retirements are creating a huge talent gap across industry, and that the young talent is just not up to the task to fill it. In fact, the point of my article is I disagree with this perspective and believe they might be very much up to the task.

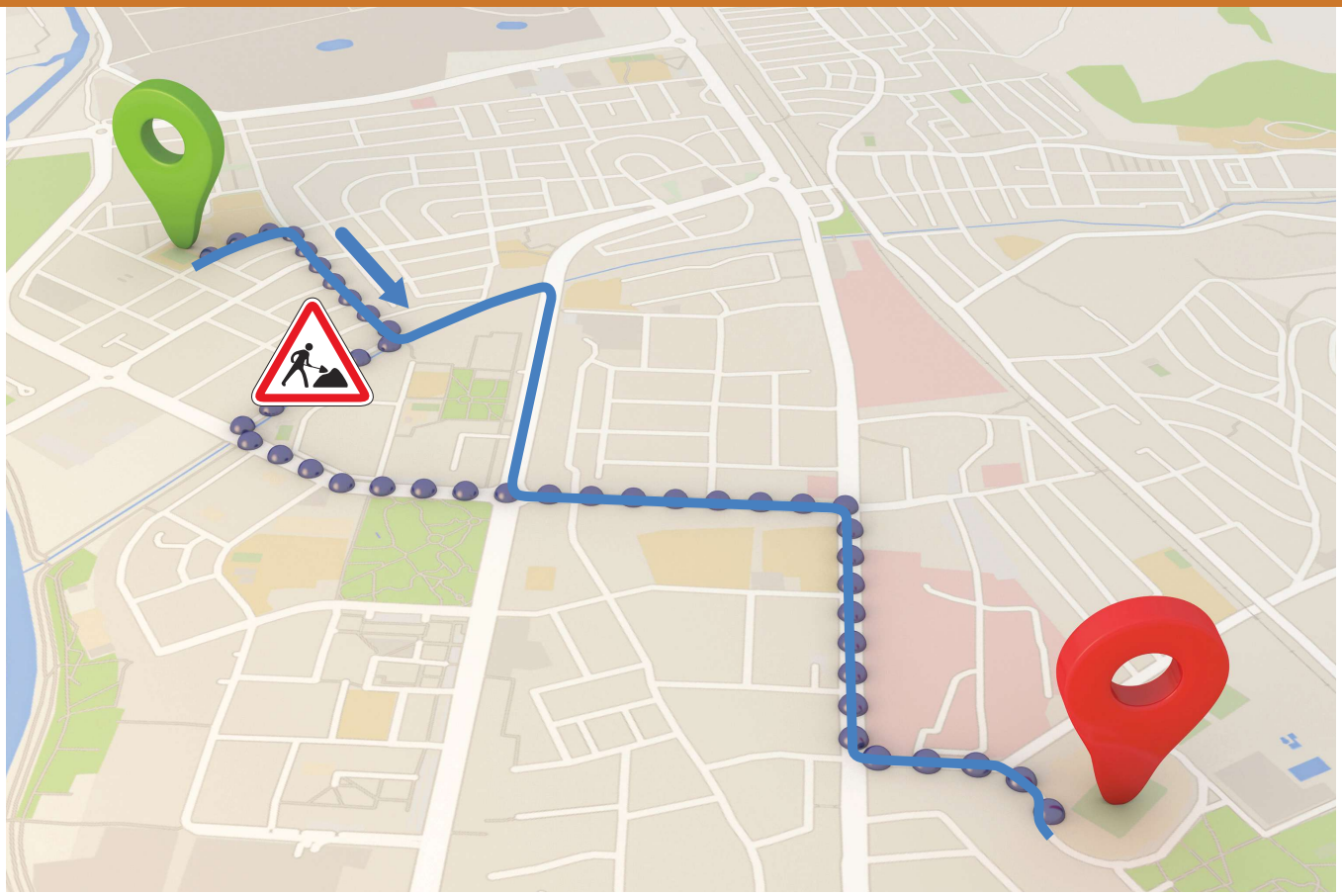
One point where I do not align with Mr. Drobny is that I do believe there are significant differences in the emerging generation that have less to do with their age and a lot to do with the actual generation. I believe this generation is the first in history to have been raised in anything close to a high-technology environment. They have a comfort and skill set with the technologies developed over the past 30 years that older professionals do not have. This skill set makes them an ideal group of people to fill the widening talent gap.

A second point is this generation on average has a somewhat unique altruistic perspective due to the era when they were raised. I believe this is because of real generational differences that are not linked to age. Coming out of World War II, the parents of the baby boomers prepared their children to gain the financial rewards that were not available in the 1930s and 1940s during depression and war. The generation today has generally been raised with different values.

I would like to thank Mr. Drobny for his insightful remarks. He makes a number of valid and important points. I just do not think we are as far apart in our perspectives as he may believe.

Peter Martin

Source: Automation.com



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Fraunhofer and German government creating Industrial Data Space

With partners from industry and support from the German federal government, Fraunhofer-Gesellschaft is launching an initiative to create a data space available to industry all over the world: the Industrial Data Space. A set of uniform standards will safeguard access and usage of the Industrial Data Space. The goal is to develop secure solutions in the face of wall-to-wall digitalization and the rapid shifts in industrial manufacturing and business processes that come with it.

"Together we want to develop tools that enable as many companies as possible to successfully participate in the digital industrial revolution. German and European industry must retain control over its own data. At the same time, small- and medium-size enterprises need a protected space in which they can share and exchange data according to rules that they themselves establish," said Johanna Wanka, Germany's Federal Minister of Education and Research.

The German federal government is being very deliberate in its support of this decentralized approach to open up the opportunities offered by Industry 4.0 to as many companies as possible. State Secretary Matthias Machnig at the Federal Ministry for Economic Affairs and Energy emphasizes, "The digitalization of industry is one of the core social and political tasks for Germany as it sets policy. Industry 4.0 is of critical importance for us as a production location, and our goal is to rigorously seize the opportunities of this fourth industrial revolution. Creating a consortium of industry and research to address issues including reference architecture, standardization, and data security, is an important step toward that goal. It shows that industry, research, and politics are all working together." ■

SCADA market to enjoy strong growth

Compared with the industrial automation hardware business, the market for supervisory control and data acquisition (SCADA) software automation products is expected to experience higher growth, IHS says. This is partly because SCADA products are mostly used in large investments in the energy business—such as in oil and gas and power facilities. SCADA's strong growth is also because more customers are becoming aware of the importance of such software, which can make production lines more efficient.

The Americas region generated the most SCADA sales revenue worldwide in 2013, and the biggest contributor was the U.S. The major factors driving growth were investments in the oil and gas markets, as well as the strong domestic consumption in the food and beverage and power markets. Europe was the second largest market for SCADA in 2013, only slightly smaller than the Americas. The market driver for Europe was its strong machinery and manufacturing industries. Russia and the Middle East were the main contributors for global oil and gas production, which is suffering from the declining price of crude oil. This price decrease is diminishing the investment opportunity. Asia-Pacific, which had been one of the fastest growing regions in the past few years, is undergoing a major deceleration. ■

Security and safety becoming big business

Anonymous threats and lone-wolf attacks, increasing fears about cybersecurity, and concerns over immigration will generate significant debate over foreign policy and how to mitigate the security risk posed by the Islamic State and other terrorist organizations. There will also be an increasing focus on how to manage the global cyberthreat, including state-sponsored attacks and the complexity of attacks on private business. Domestic policy will continue to focus on protecting borders, limiting the movement of people from war zones, and stemming illegal immigration. This issue will be a key political battleground in Europe especially, with concerns over open borders and free movement of labor. The rise of nationalism is a particular concern.

Some of the key trends in 2015 will include increased investment in technologies for improved situational awareness, both for law enforcement and the intelligence services. "Citizen safety is back at the top of the political agenda, and funding will be made available to combat technologically astute adversaries," says Steven Webb, vice president for aerospace, defense, and security at Frost & Sullivan. "Cybersecurity will also remain a key concern. Greater collaboration and information sharing between government and industry is expected as executives become increasingly aware of the threat to their business operations and shareholder value. However, the extent of collaboration and rate of investment will continue to lag behind the mounting cyberthreat to critical national infrastructure." ■

Cliff from Cheers joins AME

Actor John Ratzenberger joined the Association for Manufacturing Excellence (AME) to shine a light on the important role domestic manufacturing plays in the economy of the U.S. Ratzenberger attended AME's recent conference in Jacksonville, Fla., where he announced he would serve as honorary chair of the 2015 conference in Cincinnati, Ohio. Ratzenberger, known for his role as postman Cliff Clavin on *Cheers*, is a strong proponent of U.S. manufacturing. His Travel channel show, *John Ratzenberger's Made in America*, highlighted domestically made products. Ratzenberger recorded a message for AME members about the importance of lean practices, saying, "When you increase your knowledge, great things get done." ■

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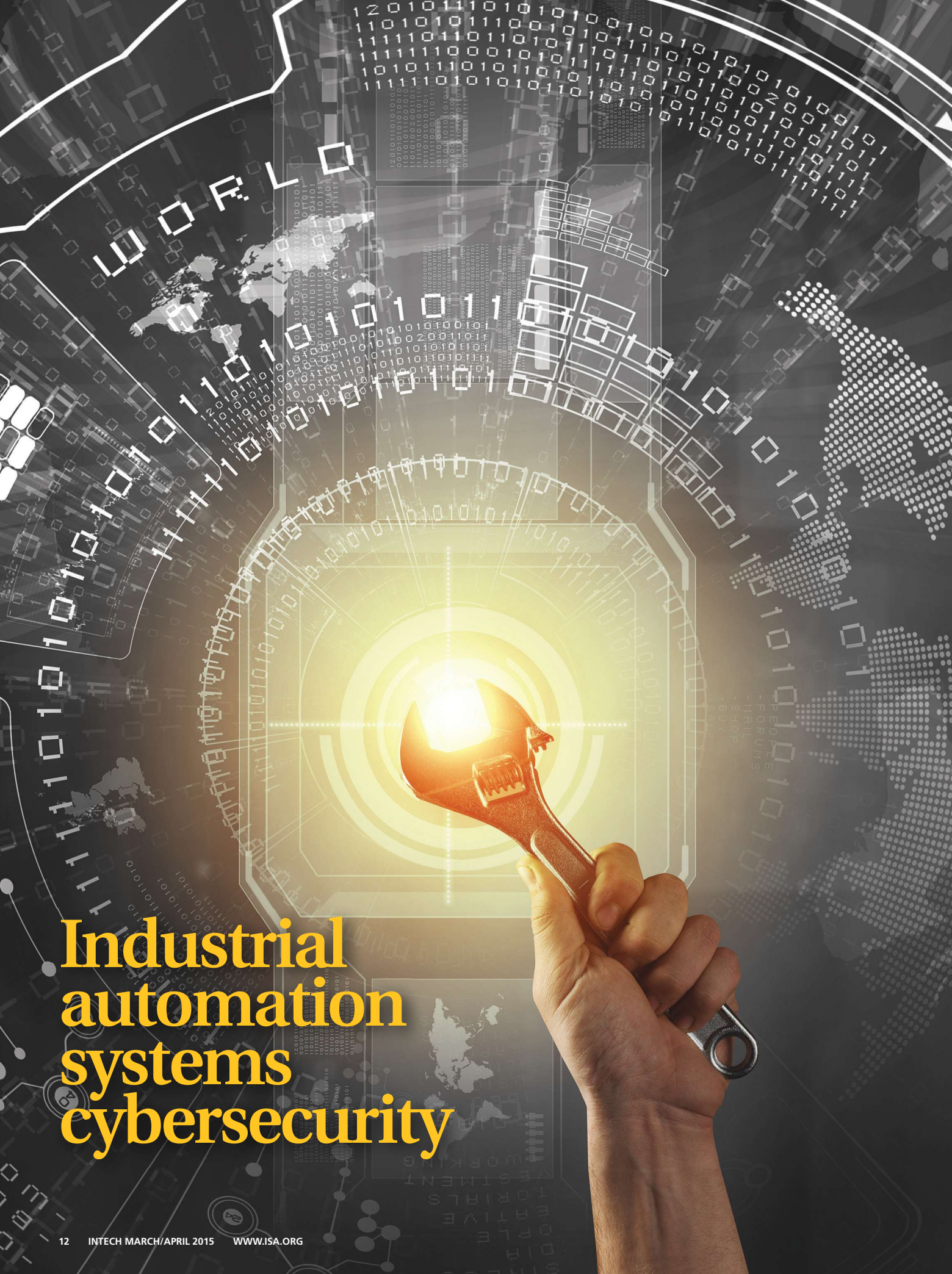

Industrial automation systems cybersecurity

FAST FORWARD

- Interconnected industrial automation systems face new security challenges such as hacking, industrial espionage, and sabotage.
- Securing these systems requires a perpetual chain of trust that spans all devices, data, and systems.
- Architectural requirements include hardened devices, secured communications, and consistent security monitoring and management.

Embedding end-to-end trust and security

By Sven Schrecker

The history of industrial automation fascinates me. Continuous innovation and new technologies have taken manufacturing processes that originated in the Industrial Age and catapulted them straight into the information age. Just as productivity seemed to be topping out, the Internet helped boost productivity and efficiency to previously unimaginable levels. Unfortunately, as industrial automation roared into today's data-driven, Internet-connected world, it sped past digital security without taking its foot off the accelerator.

Welcome to the digital age, where an anonymous hacker in some virtual landscape can throw a wrench into industrial automation systems. How do we secure these systems while still meeting the needs of corporate stakeholders? Operational technology (OT) teams still demand high resiliency and availability. Information technology (IT) teams demand interconnectivity, enterprise security, and compliance. And both of these teams must accommodate the new kids on the block: data analysts who require real-time data capture, sharing, and analysis for every decision in the business.

This article discusses the current state of industrial automation system security, the technological and organizational challenges of improving it, and a dynamic model for embedding end-to-end trust and security into industrial control systems (ICS) and supervisory control and data acquisition (SCADA) systems.

Crime and conflict have a new address

Physical break-ins and attacks on SCADA and ICS systems are largely a twentieth-century phenomenon. The overwhelming majority of attacks today are carried out by well-resourced, highly motivated attackers who are often accomplished software engineers working for cybercrime syndicates on other continents. Business competitors and nation states are the latest cyberwarfare participants, as the battleground has expanded to include manufacturing facilities, entertainment companies, and critical infrastructure. Here are a few noteworthy examples:

- The most notorious attack on an industrial automation system was in 2010, as the Stuxnet computer worm attacked industrial programmable logic controllers within an Iranian nuclear enrichment facility, subtly manipulating the feedback data of centrifuge units. This is believed to be one of the first attacks carried out by a nation state, although the source of the attack was never authoritatively identified.

A hostile takeover through data manipulation: A hypothetical example

It is a tough world out there. Unscrupulous players will use any means to improve their own prospects by harming competitors—including hacking, industrial espionage, and sabotage.

Consider this theoretical example: a major chemical conglomerate wants to take over a competitor who has no desire to be bought out. By hacking the competitor's production systems, manipulating inventory orders, or slightly altering material specifications, it could negatively affect product quality. This lowers customer satisfaction, reducing sales and driving down profitability, likely without ever being detected. The resulting shareholder dissatisfaction could create an acquisition opportunity and a favorable purchase price.

Industrial automation systems are particularly vulnerable to this attack trend because many of these systems are now Internet-connected without adequate protection. And, given the prevalence of automated systems, many daily decisions are made by machine-to-machine interactions, making them difficult to trace without proper security considerations.

Although cyberwarfare is clearly a morally bankrupt business decision, it is hard to debate its economic value.

- In December 2014, a German federal agency confirmed that a German steel plant was targeted by a malicious email that allowed hackers to cross over into the production network. The plant's controls system was compromised, preventing the furnace from being shut down. The result was the first time that "massive physical damage" to the production system was experienced; it catapults us into the new age of cyber-physical attacks with safety threats for humans.
- In December 2014, a leading industrial automation system provider patched a series of flaws in its remote terminal unit controllers used in oil and gas pipelines. The flaws included hidden functions, an authentication bypass, and hard-coded credentials, which could allow remote exploits of the devices. Although no breaches have been reported to date, the existence of vulnerabilities like this could cause extremely dire consequences.

Sadly, these types of security events continue to increase both in terms of damage and frequency. For a current list of alerts, advisories, and reported attacks, visit the Industrial Control Systems Cyber Emergency Response Team (ICS-CERT) website (https://ics-cert.us-cert.gov).

Attacking data

Data capture and analysis is today's competitive weapon—generating analytical insights that refine and optimize processes in every area of business. It is not uncommon for manufacturers to invest hundreds of millions of dollars to achieve a 10 to 20 percent efficiency increase. The efficiencies come from data-driven decisions gained through insights from customer use and demand, purchasing, supply-chain optimization, manufacturing production processes, predictive planning, and more.

By hacking and subtly manipulating data, attackers can de-optimize a company's processes without anyone even knowing.

By hacking and subtly manipulating data, attackers can de-optimize a company's processes without anyone even knowing. Even the most subtle data manipulation in any of these areas can cripple a business that is on razor-thin margins.

Security myths and misconceptions

A number of myths and misconceptions have hindered the evolution of industrial automation system security. The most common include:

- "Our OT systems are still safely air-gapped because our manufacturing line isn't connected to the Internet." This is a popular and very dangerous misconception. Since 2010, it is highly unlikely that any manufacturing control system is truly isolated. Just one user who can access the production system while logged on to the Internet, or who connects to the system with a notebook or tablet, creates security vulnerabilities. Remember the Iranian nuclear enrichment facility and the German steel plant? Enough said.
- "We are running a 20-year-old proprietary system that isn't vulnerable to modern-day attack tools and techniques." The vulnerability in legacy proprietary systems is sometimes in the communications and protocols, rather than just in the systems themselves. Security through obscurity does not work anymore. Moving from a physical world to a virtual/data-driven world powered by software poses entirely new security challenges. If there is value in data, hackers will find a way to access it.
- "Security vendors will deliver a magic box that will protect our operating technologies in the same way that firewalls and intrusion-detection systems protect our IT systems." There is no silver bullet to guarantee security throughout Internet-connected ICS systems.

Security deployment model

Establishing a perpetual chain of trust

Current client-server industrial automation systems have moved to an edge-to-cloud architecture for cost and flexibility. They have security challenges that result from today's interconnected world. Regardless of application, ensuring security begins by establishing a chain of trust between devices, data, and systems. Everything within the trusted system must be authenticated and validated to ensure trusted interoperability and integrity at every point.

Of course, availability requirements and the legacy nature of industrial automation systems add challenges. Preserving existing investments in ICS infrastructure is paramount. Therefore, a viable security model must work with both existing and new systems. In ad-

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dition, security is a dynamic process, because security needs, policies, and threat detection methods change over time. Therefore, any viable solution must be adaptable and updatable.

Foundational requirements

The embedded security deployment model establishes and ensures trusted interoperability that is essential for industrial automation interconnectivity. This model has three core requirements:

- Hardened devices (embedded security protecting "things")
- Secure communications ("things" need to talk to each other)
- Security monitoring and management (responding to changes and events)

Hardening devices

Establishing the chain of trust begins with validating the identity of the device. Previous approaches to validate device identities, such as using IP and media access control (MAC) addresses, are untrustworthy: IP addresses change routinely and can be very easily spoofed by hackers, while MAC addresses can be easily reset. Therefore, device authentication must start at the physical level—the processor within the hardware.

Device hardening may use trusted execution technology, which leverages an embedded security coprocessor (a dedicated microprocessor designed to store cryptographic keys in a tamperproof hardware container). This allows the chip itself to perform cryptographic operations such as measuring the level of trust in the boot process, an operating system, a virtual machine, or an application. A key aspect of this process is precise measurement of code, data structures, configuration, information, or anything that can be loaded into memory. Measurements consist of a cryptographic hash using a secure hashing algorithm, which allows integrity validation and detection should any measured code, configuration, or data be altered or corrupted. This is applied to software residing on the disk to determine whether or not it has been tampered with before loading the software into memory and executing it.

The chain of trust continues to be built up and verified through the complete software stack, including during the boot process, and across the entire system—even as data is encrypted and transported into the cloud.

Execution of trusted devices and data is essential given the prevalence of machine-to-machine communications driving industrial automation. For example, trusted devices can digitally sign data received by trusted industrial control sensors. Should a hacker manipulate data, the data signature will be inaccurate and be flagged by the monitoring system. In this case, the untrustworthy piece of data and the machine or sensor where it originated will be clear.

Two embedded security innovations allow communications between trusted zones of the past and the present/future: intelligent security gateways . . . and trusted execution environments.

Secure communications

Trusted transaction spaces are logical zones that allow authorized business communications. The devices must ensure the trust and integrity of data within each zone. Two embedded security innovations allow communications between trusted zones of the past and the present/future: intelligent security gateways, which enable users to securely aggregate, filter, and share data from the edge to the cloud; and trusted execution environments, which allow secure and trusted execution of application data anywhere.

Intelligent gateways: linking the past with the future

There is a reason legacy systems are so prevalent in industrial automation: they work. In fact, some have been refined for decades. New classes of intelligent gateways (some as small as two inches by two inches) are critical to extending legacy systems by connecting them to next-generation intelligent infrastructure. These gateways physically separate legacy systems, production zones, and the outside world, limiting the attack surface of an industrial automation system. The gateway can secure a device, or devices, without modifying the device in any way, making it an attractive initial security solution to create a consistent level of security within the environment.

As with any hardened device, security gateways must boot securely, be authenticated on the network, and then perform any number of security and communications tasks on behalf of the devices behind them. They can be provisioned to link trusted transaction spaces by validating integrity calculations, verifying certificates, applying cryptography, and establishing trusted communications links. Gateways can also include protocols to manage the production systems they are attached to, which can extend the life of these systems, allowing repair and updates without a physical field visit.

Trusted execution environments: security and privacy anywhere

A trusted execution environment enhances security by preventing any device from executing malicious code. It uses virtualization and encryption technologies to create secure containers for applications and data that are only accessible to approved devices. These environments are secure, trusted zones that ensure tamperproof protection of data, making data and applications invisible to third parties who may transport, store, and process sensitive information.

Even within a virtual machine that is being operated by unknown entities, the trusted execution environment can validate data authenticity and create a digital signature to attest to its integrity later. For example, production data from an industrial automation system that a cloud services provider, such as Amazon Cloud, stores and processes can be maintained securely to ensure that the data has not been secretly altered.

Security monitoring and management

There is an old axiom in IT: you cannot manage what you cannot monitor. Effective oversight of distributed industrial automation systems requires the ability to centrally manage devices through an enterprise management console, as well as the ability to monitor, collect, and analyze event information on all devices for end-to-end situational awareness of the entire system.

Enterprise security management consoles

An enterprise management console allows IT staff to manage complexity and have global visibility of highly distributed environments. The management console is where IT remotely provisions, manages, and updates software on devices, as well as defines and refines policies and pushes those policies to devices. For example, embedded devices may include whitelisting policies, which define appropriate applications, data, communications, and other functions the device is allowed to perform.

A company's enterprise management console should be tightly integrated with its security information and event monitoring (SIEM) solution and other security modules. A word of caution here: levels of integration differ considerably between vendors and security management components. A higher level of integration can greatly simplify complexity, accelerate accurate situational awareness, and reduce management time and expense. In addition, scalability becomes a critical capability for SIEMs and enterprise management consoles.

Security information and event monitoring

SIEM solutions gather, consolidate, correlate, assess, and prioritize security events from all of the managed devices that touch an industrial automation system. The SIEM combines situational and contextual awareness of all events through a process of baseline trending, anomaly detection, and alerting. Behavioral capabilities help differentiate between normal and abnormal operational patterns and refine policies to minimize false positive alerts and responses. SIEM data is also essential for conducting forensics to gain greater insight into a security incident or device failure.

Building an ecosystem

Given the distributed, interconnected nature of today's industrial automation systems, achieving end-to-end security must be a multivendor effort. To address this challenge, industry collaboration is underway, as manufacturing and critical infrastructure original equipment manufacturers (OEMs) are actively forming consortia with enterprise security vendors to ensure interoperability, set open standards, and define application programming interfaces. New systems and industrial control devices are being built secure from the ground up and designed with security technologies that ensure backward and forward compatibility.

Words of advice: Tips, tricks, and critical insights

No two businesses are the same—each has unique security infrastructures, operational technologies, and processes. Some have made considerable progress in creating converged IT/OT security solutions, while others are in the early stages. Regardless of where an organization resides on this continuum, here are some general guidelines to keep in mind.

- Establish a task force. Make sure it includes both IT and OT staff. Seek out key players in your manufacturing and industrial system controls groups, and include them in briefings and activities. Tour the factory or manufacturing facility and speak to supervisors and front-line personnel.
- Plan in phases. Target core functions that are achievable and measurable in reasonable time frames. For example, start by deploying intelligent gateways on key devices or production zones in one facility, and use that site as a pilot for event monitoring, management, and policy refinement.
- Select capable vendors who work well with others. Are potential vendors part of a proven ecosystem that includes system integrators, security experts, and manufacturing OEMs? Given the formidable complexities of securing industrial automation systems, there is no such thing as a single-vendor solution or technological silver bullet. Is security their core competency? Do they have expertise in embedded security and critical infrastructure? Lastly, can they deliver more than slideware or vision papers (i.e., do they have a reference architecture and customer references, and can they provide clear architecture designs and integration plans)?
- Insist on scalability. Make certain management and monitoring technologies scale to handle potential merger and acquisition activity, as well as what will certainly be a dramatic increase in Internet-connected devices and related security events as a company or utility grows.

Moving forward, consider how to use these core concepts to build higher levels of embedded security, secure communications, and manageability into industrial automation systems. After all, these days, no one can be too secure. ■

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RESOURCES

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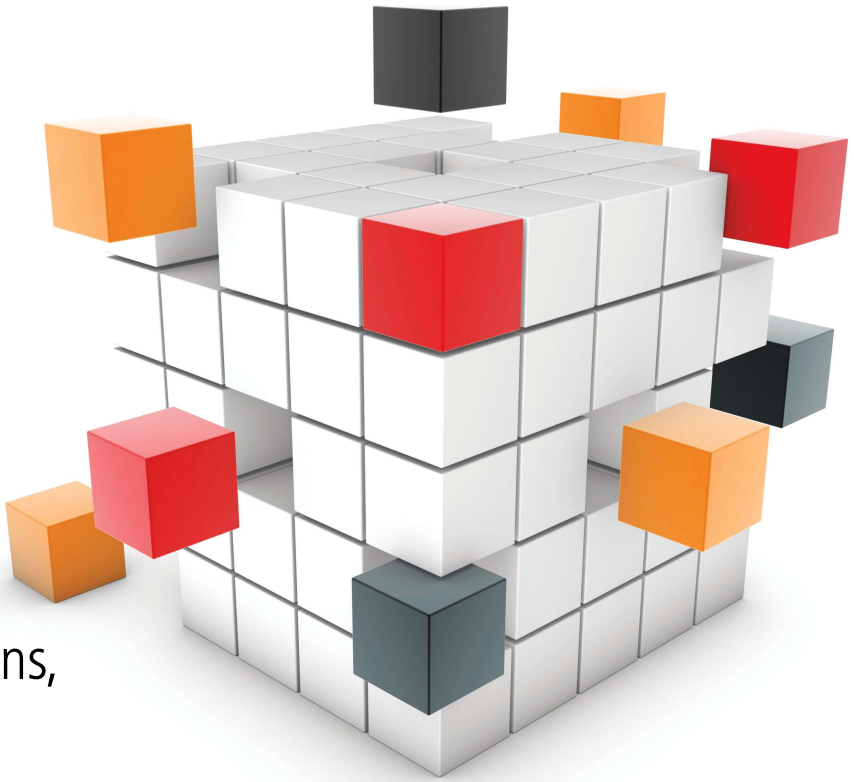
ISA-88
and modular automation



Modular systems speed and simplify new plant programming, construction, modifications, and product changes

By Marcus Tennant

Getting out the cake and candles may have been overdoing it, but ISA-88's 20th birthday was 28 February 2015. Since 1995, it has served countless manufacturers, primarily in batch process industries, while growing and evolving to cover new areas, and even spawning other standards. The first batch control standard in the ISA-88 series was approved by the American National Standards Institute (ANSI) in that year, and one could argue that the ISA-95 standards on enterprise hierarchy and ISA-106 on applying procedural and equipment models to continuous processes came out of concepts in ISA-88.

Testimonials by operating companies best represent the benefits of the ISA-88 series of standards over the years. Some benefits attributed to the standard are:

- quicker time to market as a result of reduced effort required to build plants
- ability to create recipes and equipment logic in parallel
- lower total cost of ownership for automation
- reduced training costs as engineers move between plants and process cells with different suppliers' systems and processes

Its importance continues to increase as a guide for all manufacturers, but particularly in the chemical and oil and gas industries. These and related industries are enjoying a resurgence in North America. New sources of oil and gas make the landscape far more attractive for plant upgrades and greenfield construction throughout the hydrocarbon value chain—

from upstream extraction, pipelines, and refining to bulk and specialty chemicals. In some respects, the success of this resurgence is its own greatest threat, as demand is growing rapidly for skilled engineering and construction workers. Some recent studies predict human resource shortages will impose a severe constraint in the future.

Access to cheap oil- and natural gas-based feedstocks supports growth, but keeping manufacturing sustainable in the long term depends on a variety of considerations, which is where concepts from ISA-88 fit into the picture.

FAST FORWARD

- Process equipment and even entire plants are being built off site as modules and delivered to sites as complete units to save costs and solve staffing issues.
- Although ISA-88 is now 20 years old, its guidelines are well suited to modular design and construction in new plants and upgrades.
- ISA-88 concepts are critical to creating modularized control software capable of reducing automation costs, which fits well with modular construction techniques.

Reducing life-cycle costs

Competition among global manufacturers requires low costs for all factors of production:

- plant construction
- engineering
- maintenance
- material use
- personnel
- energy

Naturally, automation is a major part of the equation. Effective control systems combined with optimal work practices support operational excellence. Due to its wide adoption, ISA-88 has had a leading role in how batch control systems are designed and implemented. ISA-88's recipe structure, the separation of product and process information, and the encouragement of design modularity have influenced most batch processes in this country, if not the world, over the past 20 years. Modularization in process industries is getting a new look in facilities, processes, and automation—and ISA-88's contributions are helping support this movement.

Start with modular construction

In years past, most plant construction was stick-built on site with carpenters, fabricators, pipe fitters, and electricians building equipment from scratch to match designs. They welded together tank sides, cut and connected piping, and added wiring and control cabinets. Small armies of skilled tradesmen and technicians had to be brought to the site for months at a time. Working conditions were not always ideal and were subject to weather interference. Even minor safety incidents were an issue, particularly as they could affect a plant's insurance rates.

Costs and labor constraints are changing the picture. Plant owners have to keep costs under control and work around skilled labor shortages, and one way to do this is by undertaking major construction projects without armies of people on site. Such constraints push companies to be more creative and do more with fewer resources. One option, which is growing in popularity, is building more of the plant equipment as modules in specialized facilities rather than on site. These modules are delivered to the construction site ready to be interconnected.

Modularization has been going on for decades, but the variety of equipment being handled this way is growing. The more manufacturers can do in a specialized environment with ideal tools and the most highly trained people, the easier it is to control costs and ensure the highest possible quality and safety. Process plants are embracing modularization in new ways and are gaining comfort with the practice.

Equipment delivered to the site in modules ready for installation can be dropped on a foundation designed to receive it, with piping and wiring in place and ready to connect. Modularization saves time because equipment can be built off site before all permits are received for a new facility. When final approval comes in, much of the equipment will be ready to install immediately, rather than having to wait to begin the construction process. Any module not too large to be shipped to the site can be built elsewhere and brought in as close to complete as possible. This can go as far as an entire plant or production unit.

Modularity beyond construction

The idea of modular equipment and construction is easy to understand, but ISA-88 is not really about equipment. However, it does go hand-in-glove with the modular movement by providing principles for design following the fundamental ISA-88 model: the physical model and the procedural control model (figure 1). ISA-88 offers guidelines to help build logical,

The ISA-88 physical model defines the different levels of equipment structure

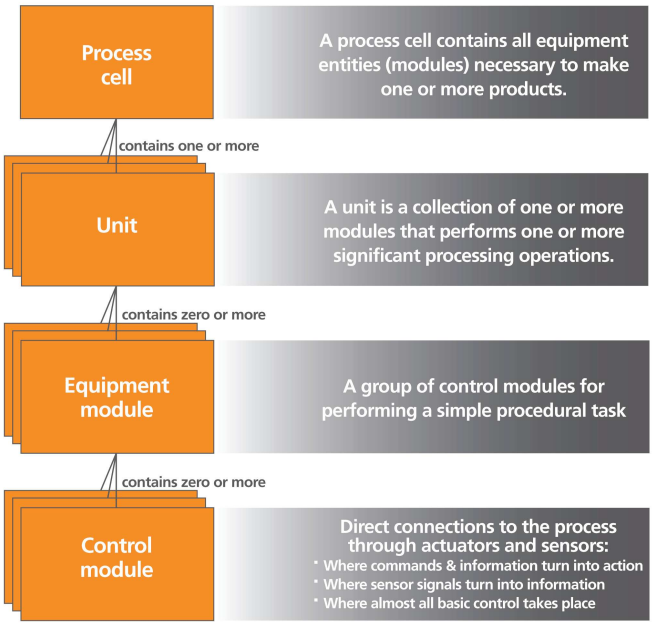


Figure 1. Understanding equipment structure is critical in an environment where hardware from a variety of sources has to work together seamlessly. In the 2010 update of ISA-88, the definition of a process cell was changed to allow it to directly contain either an equipment or control module, and a unit can directly contain a control module.

internally consistent manufacturing systems by helping users understand concepts of modularity. Just as modular equipment is a way to save money, so is modular manufacturing.

To understand modularity, consider the opposite approach. Old-fashioned batch manufacturing involves following a specific recipe using specific equipment. For example, to make 100 gallons of product X, begin by putting 50 gallons of feedstock 1 and 50 gallons of feedstock 2 into tank #100 and inject steam while agitating until the temperature reaches 160° F.

The recipe, procedures, and equipment are linked together and cannot be easily separated. This works very well if the company wants to make only product X in exactly the same way in the same amount with the same feedstocks using the same equipment indefinitely. But changes or deviations to add products or make alternate versions are complicated, because it is difficult to separate the various production elements from the final product.

ISA-88 helps users understand how the steps involved in making a product can be pulled apart and made into abstract modules (figure 2). One of the first steps is to separate the recipe from its equipment. The recipe simply describes what raw materials go into the final product and what has to happen to them in basic terms. A manufacturer can look at the recipe and determine if a given set of equipment or a process unit has the capability to perform those functions. For example, does tank #200 have a steam injector? An agitator? Is there any reason we cannot make product X using this process unit?

Equipment needs to be thought of conceptually as modular, too, not just from a manufacturing standpoint, but also in terms of configuration and arrangement. With ISA-88, a process is considered in terms of modules, with dedicated control logic for each. Each module and its associated control code perform process tasks, but these are not product specific. A process cell can include multiple modular units. When equipment is modular, it can be set up to satisfy the requirements of numerous recipes or manufacturing campaigns by mixing and matching modules as required.

With modules, a recipe is product specific but not equipment specific. A recipe sets the order in which process modules should execute and how they should be linked to make the desired product, but operators can choose any equipment capable of fulfilling those specifications.

Along with equipment and recipes, control code and related automation infrastructure is separated into modules connected to the associated equipment (figure 3). If a vessel has a heating element module, the control code to regulate the heating element should be part of the module.

Modularization of a unit

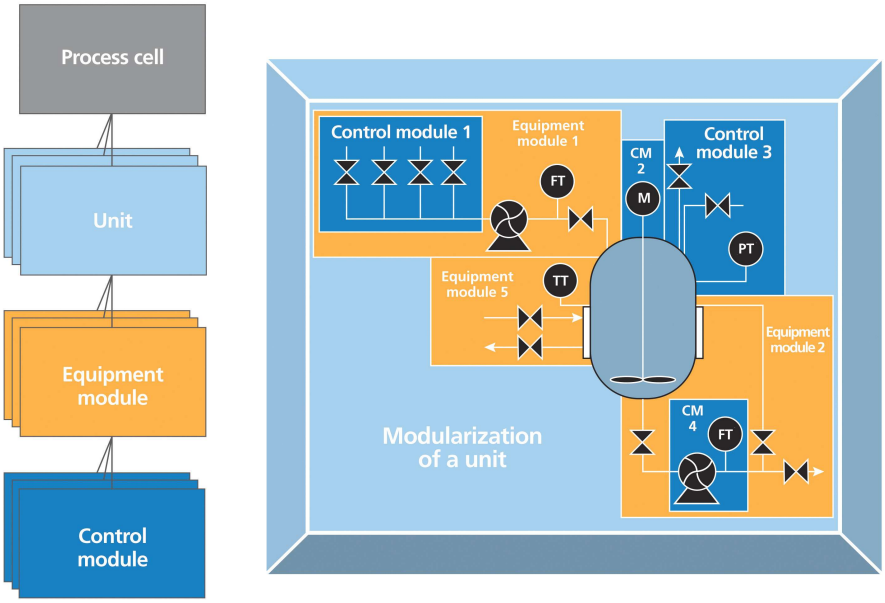


Figure 2. Actual equipment has to be separated into different levels following the physical model definitions.

TRUE STORY

When an overhead crane accidentally dropped a steel pipe 15 feet onto an IceStation, workers were relieved to find the pipe had not been damaged.

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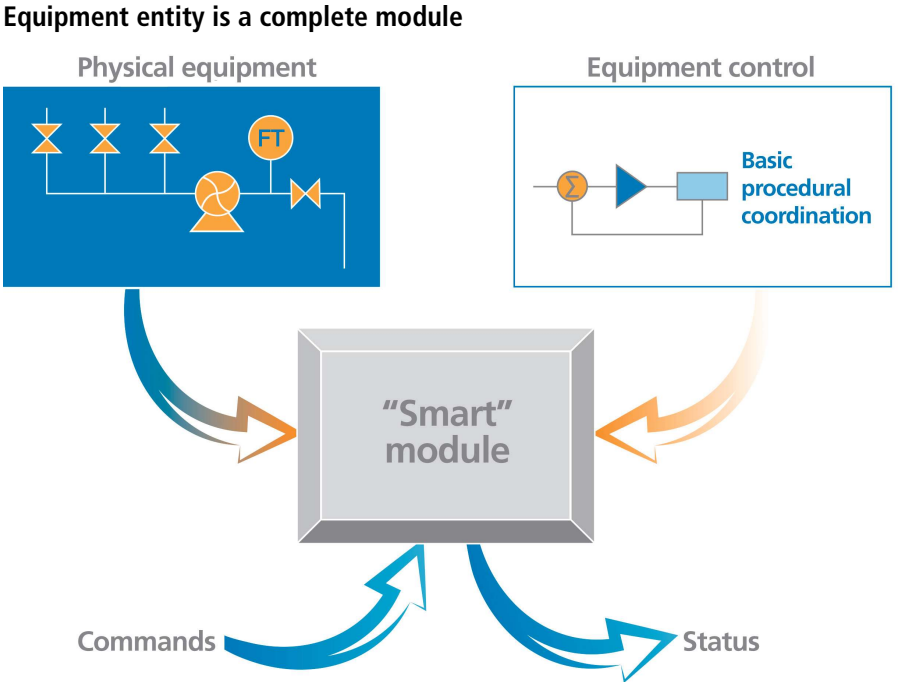

Figure 3. Equipment entities are formed by combining equipment control and physical equipment. They can accept commands and send out status.

Do not forget the software

Software for control and procedure management can also be modularized, and ISA-88 makes critical contributions to realizing this objective by providing common batch control language and by defining models and terminology (figure 4). In fact, its original function was as an engineering tool to fill critical gaps for engineers trying to create structure for procedural control, process management, and interaction with the business level to make everything work efficiently and effectively.

The standard has basic functions for addressing modularity issues:

- consistent terminology
- models to define tested structures
- abstraction to support manufacturing and provide design freedom
- standardization concepts for software producers

ISA-88 did not invent the concept of creating reusable control code, but it has mechanisms—language and organizational models—to make implementation far easier and more practical. Its structures allow engineers to define elements as simple as "stir" and "heat" along with more complex concepts, so code creators can structure programming by linking modular blocks of code, rather than by writing a monolithic program.

How much can modularity encompass?

Entire plants are being built today around modular concepts. Although there is still much to learn, there have been enough positive results to prove the concepts. ISA-88 has helped avoid many interoperability and plug-in issues common when multiple suppliers in different locations work on a single project.

The biggest constraints to modularization relate to physical shipping limitations. There are inescapable realities of tank and vessel sizes to support specific production levels. Anything unable to fit on a truck, railcar, or barge has to be fabricated on site, but for most plants, these turn out to be a very small number of items. Generally such size considerations only involve plants dedicated to refining or large volume petrochemical production, leaving the components of a typical food and beverage, pharmaceutical, or chemical manufacturing facility largely within the constraints of module shipping size.

Plant designs using module concepts can be scaled up or down and reconfigured or adjusted as needed to satisfy specific customer requirements. Modules that do not need to change can remain intact, so alterations only need to be made to specific modules, or to the manner in which these modules interact.

Extending the reach of ISA-88

As mentioned, over ISA-88's 20-year life span, other standards have emerged by extending its basic concepts. Here are some examples of current ISA standard development work influenced by ISA-88.

ISA-101 on human-machine interfaces (HMIs), expected to be published by ISA in September 2015, considers how menu hierarchies, navigation, graphics, color use, and other dynamic elements can help tie together elements from a variety of suppliers to create a seamless interface for operators. Modularization on a large scale can be greatly simplified with this type of integration guidance. Additionally, ISA-88 affects development of HMI graphics to support a modular structure. The physical model determines how graphics are built and organized into a hierarchy.

ISA-106 on procedural automation for continuous processes draws from lessons learned using ISA-88 in batch applications over the years. Because procedures (e.g., startup, shutdown, grade change) used with continuous processes are much like batch processes, many of the same concepts apply and can be used to smooth over these critical transition periods when so many safety incidents and production interruptions occur.

The ISA-88 Part 5 standard currently in development defines a reference model for modular equipment control within equipment entities to address the control ascribed to equipment phases and equipment modules. The reference model applies to equipment

The ISA-88 Procedure Control Model

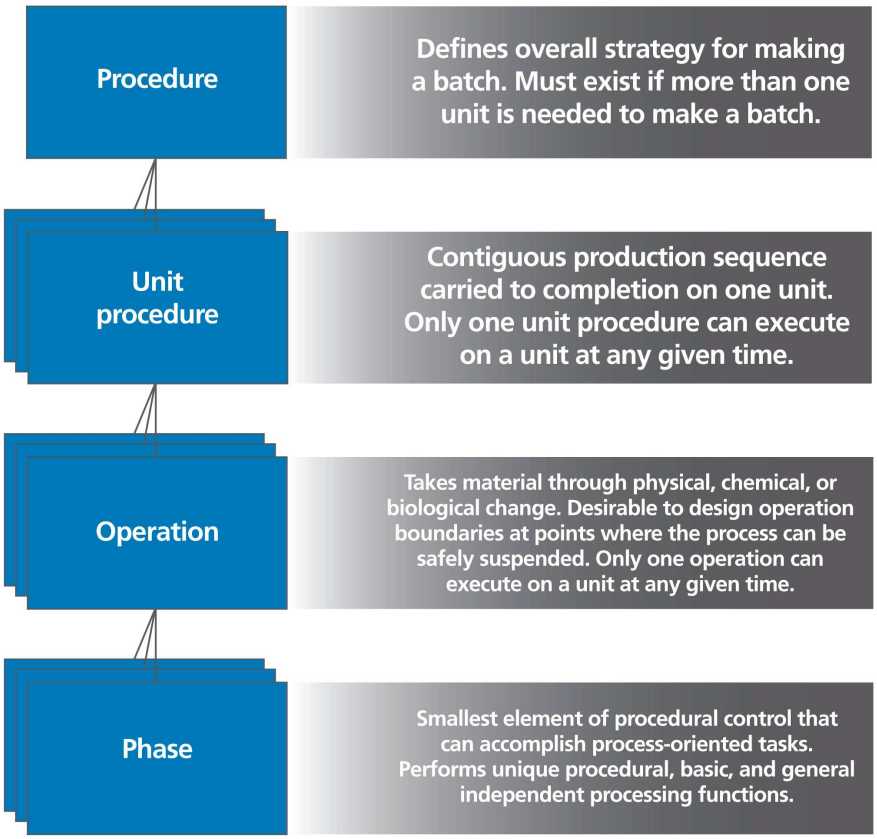

Figure 4. Just as equipment needs to be separated into its functional levels, so must procedures following common language and terminology.

that may be used in batch processes and on other types of production and packaging processes. It defines:

- a template for reusable modules and standard terminology
- a template for command and control functionality within and among modules
- a method to describe and identify each module
- a method for exchanging module definitions among interested parties

ISA-18.2 Technical Report 7 on alarm management for packaged systems (expected to be published by the end of 2015) looks at the practical side of incorporating modular equipment, such as prepackaged skids, into larger control system and alarm management strategies. The importance of this will grow as the variety of prepackaged and modular systems increases, and as they become more common in all types of plants. Using this standard involves defining different types of system architectures and defining how they are integrated into control systems. It also includes aspects of HMI integration including how alarm information is passed along and displayed for operators. Larger life-cycle management issues connected to alarms and other aspects of control integration are also being considered.

ISA-88 is still being applied in new environments, and there is more to be learned about the implementation of modular manufacturing in many forms. The standard has even entered into the world of discrete and continuous manufacturing, because many of the same concepts apply. ISA-88's application has expanded over the years, and this will continue as manufacturing adopts more and more modularization. ■

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Marcus Tennant (marcus.tennant@us.yokogawa.com) has been with Yokogawa Corporation since December 2008 as a senior principal technology specialist. Before Yokogawa, Tennant worked at Rockwell Automation for 10 years as a product manager and application engineer. Before that, he was with Morton International for 10 years, holding various positions in process development, project engineering, and quality assurance and with Jones-Blair Company for five years as a research and development chemist and process engineer. Tennant has a B.S. in chemical engineering from Michigan State University and an M.S. in operations and technology management from the Stuart School of Business at Illinois Institute of Technology. He has been a member of AIChE since 1984 and is also a member of ACS and ISA.

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"Batch integration: An enterprising story"
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PLM as part of a connected future

Product life-cycle management and automation are being drawn together

By Simon Hailstone

The worlds of product life-cycle management (PLM) and automation are being drawn together by a series of technological developments that are beginning to have far-reaching effects, not only in manufacturing, but across all industries.

For PLM vendors, this combination is manifesting itself in an upsurge of research and development activity, acquisition, and partnering. For those in industry considering future investment, from both technology and business improvement perspectives, the decision making just got harder. The driving force is a combination of the elements of hardware, "smart," and cloud.

Hardware and devices

Additive manufacturing and printed components are set to revolutionize how products are developed and delivered. Fully defined 3-D models give the manufacturing definition directly to the point of production. New industrial devices are becoming more powerful and capable, enabling direct communication with level 4 and 5 business systems. Scanners and autonomous vehicles provide new ways to capture the as-built, as-produced, as-operated environment.

"Smart"

Products, buildings, and even cities are becoming smarter and smarter as automation devices and networks provide comprehensive interconnectivity. However, continuous improvement in capability requires connection to the product and asset definition and a way of managing the closed loop process.

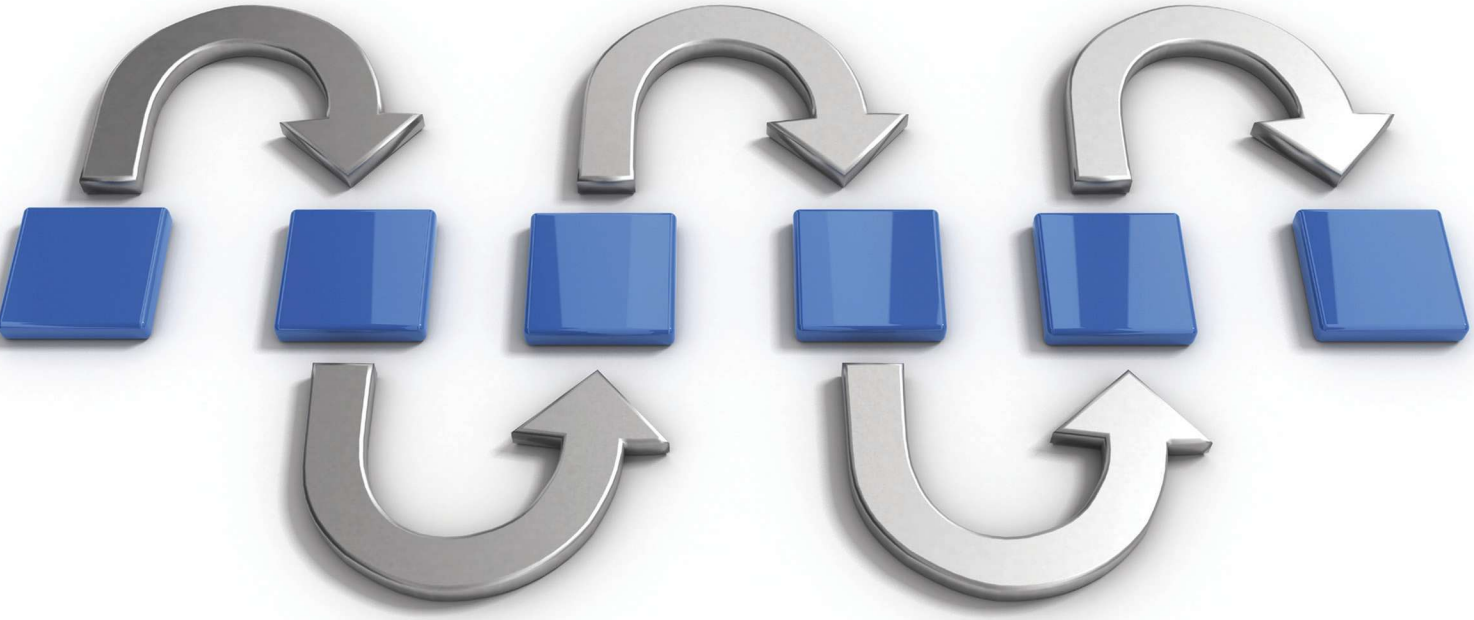
Cloud

Proliferating mobile technologies enable access to information anywhere. Cloud-based infrastructure may be a more cost-effective mechanism for smaller companies to fully participate in a highly connected environment. These influences are part of a verticalization across industries, from service to hardware, that affects how products are defined, produced, installed, and serviced. In other words, the trend is to cover the complete life cycle, and that is why PLM is an essential part of this process.

Here we examine how PLM is coping with these drivers and the changes occurring across multiple industries. The article uses the terms *product, service, facility,* and *asset* somewhat interchangeably to reflect the fact that end-to-end life-cycle management applies to all of them, although the routes to operation and the time frames may differ widely.

Cross-industry influences

For some industries, managing the complete life cycle has been a way of life for a long time due to the long-term nature of the assets. These were usually highly capital-intensive and regulated industries, such as nuclear power, marine, and civil construction.



The construction, process, and mining industries have been monitoring, collecting, and processing operational data for years. Facilities, asset management, and location tracking solutions capture significant volumes of data daily. Does PLM technology have a role in this future connected world? In this context, PLM is a "wrapper" around the life cycle of individual assets. It can provide the definition and operational parameters of individual equipment items—whether they are pumps and actuators; elevators; heating, ventilation, and conditioning; or heavy equipment—and respond to in-service issues. Performance data—whether it is flow and temperature in process plants, structural deformation of structures, or environmental control in inhabited spaces—can be processed for more than just short-term corrective action. It can be fed back for resimulation as part of long-term continuous improvement and for data for next-generation design.

One consequence of the continuous quest for more intensive asset use in these industries is the adoption of mobile technology. This has historically focused on activities such as task management, operational data, and fault recording and reporting. The trend is toward delivering live information as 3-D models, animations, and service data. This kind of delivery is common in some manufacturing industries with production instructions, simulations, inspection information, and exception reporting being delivered and processed directly at the point of production via PLM.

As the building and construction industry moves more toward modular, fabricated structures and even "printed" buildings, the role of PLM may increase. The processes involved are more in the realms of traditional manufacturing and assembly than of construction. China is using modular construction extensively, and it has grown in the U.S. Couple this with a proliferation of smart sensors for buildings—to optimize not just utility usage, but also performance of equipment and finishes—and the ability to reconfigure spaces and PLM starts to become a key capability in this sector. The worlds of construction and manufacturing are colliding in the sense that information about products, the processes and machinery that produce them, the facilities involved, and the supply and delivery networks that support them are coming together as never before.

But we must be careful not to get carried away with the notion of PLM as an all-encompassing technology. Many other enterprise environments already provide much of this capability, from traditional facilities and asset management solutions to the rapidly developing building information management (BIM) sector. As physical environments become more connected, the underlying supporting infrastructures need to interact and connect

FAST FORWARD

- PLM and automation are being drawn together, which will have far-reaching effects across all industries.
- The driving force is a combination of the elements of hardware, "smart," and cloud.
- R&D, acquisition, and partnering are all increasing for PLM vendors.

seamlessly and reliably. This is as true in the PLM world as it is in the automation world, where established protocols and standards (e.g., ISA-95, OPC UA, BACNet) are also being challenged to support new levels of connectivity.

The PLM contribution

To put this in context, it is worth taking a step back to look at some core capabilities associated with PLM. The scope is potentially very wide, so we will focus on two key areas particularly relevant to this discussion: product definition management and configuration management.

PLM product definition encompasses requirements, systems models, 3-D models, tests, instructions, process plans, tooling, quality metrics, service information, and packaging. These areas would traditionally have been in the form of documentation, but are increasingly captured as part of a complete virtual definition. Product definition also includes the definition of product structures (bills of material) and, critically, the process trail that led to the definition. This latter capability is vitally important when considering the potential increase in feedback from both production and in-service monitoring of smart products.

However, the scope of product definition is changing. PLM grew out of the discrete manufacturing arena—hardware focused and engineering centric with classical bills of material. Now PLM is found in service industries like telecoms, finance, fashion, and pharmaceuticals. It has to manage products with hard, soft, electrical, and electronic components. Even the hard components are changing with the adoption of composites and other new materials. This brings changes to both product definition and production processes and equipment.

Increasingly software is the key value benefit for finished products, allowing incremental improvements to products in service. This, combined with in-service monitoring, is moving the definition of products toward service provision where the physical item is only part of the "product." Manufacturers can learn from service industries that have deployed PLM effectively to define and support a portfolio of service offerings.

As the move toward service accelerates, how will it affect providers of the wealth of monitored data from in-service products? Will we see a significant rise in recalls as both producers and consumers have access to more defect and risk-related information? It is clear that the full audit trail from definition to delivery will become increasingly important.

Configuration management is the second key capability. This is not just version control, it is the management of multiple complex configurations of multiple product lines, maintaining not just bill of material definitions but all the associated definition data. As products become more customizable, managing this complexity will become more important. In particular, as software updates become part of the service life cycle, products must be defined to enable future modifications in a way that enables sufficient scope for change. Of course this is not a new problem. More than 25 years ago, a major provider of automation systems stated that coordinating the configurations of hardware and firmware releases was a major issue resulting in significant warranty claims. We do have better systems in place now to help prevent this, but in the future it will be orders of magnitude more complex.

Coordinating all the information involved in developing, manufacturing and building, and servicing these mixed technologies, mixed material products (produced locally and delivered in an increasingly customized world) requires configuration management capabilities of considerable force coupled with rigorous traceability.

Broader visions

How are solution providers in the lifecycle management space encompassing a broader scope?

Within the primary PLM providers there have been notable developments. Siemens has been forging the connections between PLM and industrial automation ever since it acquired Unigraphics in 2007. Dassault Systèmes' acquisition of Apriso puts it in the center of this coming together of worlds. Now under the Delmia brand, Apriso is part of a wider capability for virtual simulation of factory and production planning and commissioning and operations management. PTC has opted for partnership rather than acquisition to provide verticalization within manufacturing, with an intriguing tie up with GE Intelligent platforms. However, PTC's focus is emerging in the service part of the life cycle with its Internet of Things (IoT) capabilities courtesy of Axeda and ThingWorx. Accomplishing the goal of Industry 4.0 and the IoT is creating the need for more sophisticated and integrated PLM systems in the industrial automation market.

PLM grew out of the discrete manufacturing arena—hardware focused and engineering centric with classical bills of material. Now PLM is found in service industries like telecoms, finance, fashion, and pharmaceuticals.

Cross-industry developments require knowledge and experience outside of the historically restricted view of PLM. Service organizations, such as IBM, HP, and Accenture, are gearing up for this, with Accenture recently extending its PLM services capabilities. Alongside this there are signs that other industries and third-party providers are recognizing the need to bring together definitional and operational data. Rand Worldwide has merged its Imaginit Technologies and facilities management (FM) divisions, bringing together the worlds of BIM and FM.

Although not necessarily providing PLM in the traditional sense, a number of providers in adjacent industries

are extending their capabilities in ways that will overlap and interact with PLM. Bentley Systems has been developing its presence in infrastructure asset management to complement its BIM and project control capabilities. It has hooked up with Siemens for complementary capabilities in constructing and operating factory facilities, perhaps signalling a move toward further cross-industry cooperation. Autodesk has a cloud-based PLM solution and has partnered with cloud enterprise resource planning (ERP) provider NetSuite, demonstrating the potential for open interaction across enterprise platforms. Autodesk is also the major provider of solutions for architecture, engineering, and construction (such as BIM). With its concerted moves toward cloud deployment, it is positioned to support cross-industry life cycles.

The cloud also gives new companies the opportunity to develop PLM-like solutions rapidly and provide services without large-scale infrastructure investment. These solutions have tended to focus on definition phases of the life cycle, but some interesting trends are appearing. First is the acquisition of GrabCAD by 3-D printing company Stratasys. GrabCAD cannot strictly be considered a full PLM system. However, the close connection between product definition and product manufacture is important, because it is going to change the notion of industrial production at some point. A second area is the coming together of core PLM capabilities, such as workflow and configuration management, to support application development for IoT. Companies such as Solair Srl. based in Italy are starting to provide cloud-based IoT application platforms that link product in-service data with product definition data.

Avoiding the silos of the future

Life-cycle management environments, supported by enterprise software, have been chipping away for some time at the organizational and functional silos of companies and their extended value chains. But the technologies that have aided this are themselves in danger of creating their own silos. As vendors jockey for position, the danger lies in creating new silos. These are not the traditional functional and organizational silos of engineering, manufacturing, and supply; software, hardware, and electronics; architecture, engineering, and construction; and utility production and network planning, but these are data silos that hold the potential for enormous value.

Is there one answer as to where this data should reside? The notion of a "single source of truth" is often mooted in the PLM world, but what does this really mean in practice? How can this work in a future yottabyte ($1{,}000^8$ bytes) world? This is not a job for a single source of truth. This requires a highly connected solution stack that includes at least PLM, ERP, and manufacturing execution system capabilities in manufacturing and extends to facilities and asset management, BIM, and other industry-specific environments.

The answer surely must be openness driven by protocols, standards, and defined architectures. In the same way that protocols are essential to communication between automation components, similar protocols are required to connect the data service platforms that are currently delineated in the realms of enterprise systems. But with statements such as "Data is the new oil," the stakes may be too high to allow for such openness. Without it the smart, connected future and everything that goes with it will not achieve its true potential. ■

ABOUT THE AUTHOR

Simon Hailstone (Simon.Hailstone@cambashi.com) has more than 20 years of experience in the engineering software industry. Before joining Cambashi in 2008, he held a variety of positions ranging from CAD/CAM R&D, software development, and database deployment through sales and technical account management to project management and process change. Hailstone has particular knowledge of PLM business transformation needs and in-depth knowledge of the supporting technologies.

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Realizing more value from automation projects

By Peter G. Martin, Ph.D.

Change our approach to performance measurement

The industrial automation industry is largely a project-centric business. Over the past three decades automation suppliers, system integrators, engineering procurement contractors, and industrial automation users have become extremely proficient at executing automation projects.

Unfortunately, in numerous interviews in which I have participated with industrial company executives, many indicated that although they believe automation should add business value to their operations, they are not convinced that it really does. Many of these executives have commented that they see huge return-on-investment (ROI) projections for automation projects before selecting and executing the projects, but seldom see any actual ROI data upon completion. As a result, they have become skeptical about the value of automation projects, and in some cases are actually redirecting available capital to other areas.

This skepticism has been a reality for many years. Perhaps the best proof of this is the continual price pressure in the automation marketplace, which has made automation technologies commodity products. This degree of commoditization would not have been as significant if executives truly believed automation projects delivered high levels of business value.

Barriers to automation project value

Most automation professionals I have talked to are convinced that automation technologies, effectively applied, can add significant business value. But that does not seem to be the prevailing perspective of executives. It is critically important to the success of industrial automation projects that the business value of automation solutions is made clear and discernable and that automation investments are made in the areas of greatest potential value improvements. Driving this transition requires identifying and overcoming the barriers to measurable business value improvement.

There are three critical barriers to realizing the potential business value generation from automation projects:

- Replacement automation
- Project team performance metrics
- Lack of appropriate business measures

All three of these barriers are based on traditional business processes associated with selecting and executing automation projects. Fairly minor modifications to these traditional business processes will cause greater discernable value from automation. This should help change the prevailing executive perspectives that automation projects do not produce value.

Replacement automation

There are two basic classes of automation projects: greenfield and brownfield. Greenfield projects are projects in new plants involving new automation systems. The installation and startup of the automation systems are often on the critical path of greenfield projects. Therefore the primary objective for greenfield projects is getting the automation systems installed and running correctly in the necessary time frame to get the plant running on schedule. As a result, the business value perspective proposition for automation systems in greenfield operations is essentially viewed as an enabler to getting the plant producing and delivering profits. During the 1970s when a large percentage of the automation projects were greenfield, there was very little question about the value of automation.

Brownfield projects, on the other hand, are projects in existing operations. They typically involve the installation of new automation technologies to upgrade or replace existing automation systems. Over the past decade, the vast quantity of automation projects have been brownfield projects. As a result, the impressions of executives regarding automation projects have been highly affected by brownfield projects.

Unfortunately, the commonly accepted capital budget processes associated with the application, selection, and execution of brownfield projects have severely limited the actual value of automation projects. One reason for this is a phenomenon called "replacement automation." Many brownfield automation project opportunities are first identified by plant engineering and operations teams motivated by an installed automation solution that is aging, with an increasing frequency of failures and difficult-to-procure spare parts. Team members decide to request capital to replace the aging system. To develop the request, they need some idea of the cost of replacing the system, so they develop an initial specification for the new system. They technically define the exact system that is already installed and to be replaced. This specification is sent to a few suppliers to establish an estimated replacement cost, and this is the cost that is typically used to request capital funding. This means that the initial project specification is for the exact functionality of the system being replaced, and the amount requested is what it costs to execute a project for that exact functional replacement.

FAST FORWARD

- Effectively applied automation technologies add significant business value.
- Automation is one of the highest return investments for industrial companies.
- Real-time automation value indicators make automation project value tangible for management.

KEEP IT ALL UNDER CONTROL

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As our most experienced engineers and operators approach retirement age, the skills gap is widening, particularly in the automation and IT fields. Knowledge transfer for your legacy systems is exceedingly difficult when you don't have anyone to transfer it to. And as your operations expand, it becomes ever more challenging to monitor production and efficiency in a meaningful way.

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Updates and patches are often deprioritized in favor of immediate needs, leaving your system vulnerable to attack. Plus, these procedures have to be performed in a very systematic way. A remote management solution can monitor update release schedules and test updates and patches prior to implementation. It can also weigh the risks against the rewards and communicate the findings to plant personnel.

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harden your system against cyberattacks by keeping your system up to date at all times. They also alert plant operators to potential issues so they can take action before a system shutdown occurs. This helps maximize production while freeing up your staff to focus on operations.

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To secure capital funding, the plant engineering and operational team estimates the value the new automation system will provide, typically in the form of an estimated ROI. Because the new system is a functional replacement of the existing system, the only real data the team has is spare parts and maintenance cost savings data, which is seldom enough to justify the required level of ROI to get a capital project approved. Therefore the team estimates the ROI based on what it determines to be the threshold level for project approvals for the company. The good news is that the actual ROI is seldom calculated after project execution, so whatever ROI they estimate is pretty safe for the team.

If the capital project is approved, the specification and the project budget given to the project team is for an exact replacement of the installed system. Even though the new system may have many additional features and functions that could add value the old system did not have, those features and functions are typically not included in the project scope. This is because the plant engineering and operations team was likely not aware of the available capabilities when it initiated the project. As a result, the cost to implement those features and functions was not included in the project costs. Therefore the project is typically for an exact functional replacement of an existing system. Unfortunately, replacing old technology with new technology performing the exact same functions seldom has significant incremental value.

The "replacement automation" approach produces a number of projects that may be well implemented, but provide little incremental business value. This means that for many automation projects the executive perspective that automation projects do not add much value is correct. It is important to point out that this does not mean that the new automation systems cannot provide significant value improvement—they often can. But realizing the incremental business value improvement depends on using some of the new capabilities not available or not utilized in the installed systems, and the project budget does not support that work.

Changes to current processes are required, but the increased returns from the automation investment should be great enough to make the changes very worthwhile.

It is interesting that project teams are often in the position of selecting, or at least influencing, the selection of the automation system. This may result in automation selection criteria heavily weighted on "ease of use." That is, the project teams put heavy emphasis on selecting technologies that make on-time, on-budget project delivery more likely. Ease of use can be attained in a number of ways, including reducing the functionality of the system. Reduced functionality may imply less capability to drive performance improvements. This is not to say that automation systems should not be easy to use—they certainly should be as easy to implement and use as possible. But reducing value-adding functionality to attain ease of use may not be the best trade-off for long-term performance.

In discussions with project team members on the topic of unused functionality that may drive significant business value improvements, many expressed frustration that they could not take advantage of the functionality during the project. But they also indicated that they were optimistic that this incremental, unutilized functionality might be used later, after the project was complete and the system started up. In my experience, very little, if any, of this incremental functionality is ever implemented. Plant engineering teams lack expertise, and plant engineers have high activity levels, leaving no time to work on these potentially beneficial opportunities. The result is that the utilized functionality of automation systems changes very little over the system's life cycle, and the potential incremental business value improvements are not realized.

Project team performance metrics

One potential solution to the replacement automation challenge is adding a key project team performance measure on the incremental business value provided by the automation solution. This sounds like a simple fix to this issue, but it really is not simple at all. It requires rethinking the entire project cycle. Project team members might need an opportunity to influence the specification to drive value-adding functionality before project costs and schedules are nailed down. Time needs to be allocated to project cycles so project teams can add value in this way. Project managers need new mechanisms to manage scope creep effectively while allowing appropriate, value-adding func-

tional enhancements. Changes to current processes are required, but the increased returns from the automation investment should be great enough to make the changes very worthwhile.

Although the replacement automation approach results in project specifications that add little or no value, there is often a high level of talent on project teams. It might be expected that the project teams could use this talent to add incremental value. Unfortunately this is seldom, if ever, the case. The problem is not the talent level on project teams, but how the performance of project teams is measured.

In most industrial organizations, the primary performance measures of project teams are on-time and on-budget delivery of the projects. Notice that the actual incremental business value delivered by the project is typically not a key measure of project team performance. In fact, in many of the projects I have been involved with over the past three decades, it is not a project team performance measure at all. Project budgets and project schedules are set up to align with the project delivery according to the scope defined in the project specification. Therefore the time and cost allocated to the project team are set up to deliver the project as specified and nothing more. Because project team managers are extremely cognizant of their performance measures, they tend to carefully manage the projects to ensure there is no scope creep that might negatively affect schedule or cost. This means that the functionality in the project specifications, which is often defined on a replacement automation basis, is exactly what is delivered by project teams. Unfortunately this leads to automation projects that provide little or no incremental business value.

Lack of appropriate business measures

The third barrier to driving improved business value from automation projects is the lack of appropriate business measures to actually measure the incremental business value delivered. The cost accounting systems installed in most industrial operations do not collect data in the necessary time frame or with the necessary functional scope to effectively and accurately measure the business value impact of brownfield automation projects. The time frames of today's cost accounting systems are typically monthly, weekly, or even daily, but they do not have the real-time perspectives necessary to discern the effect of most automation projects.

Additionally, the functional scope of most cost accounting systems is an entire plant. Most brownfield automation projects are focused on a much smaller domain. An advanced process control project, for example, may reduce the energy cost to a single process unit, which is one of 40 in the plant, by $250,000 per month while maintaining production throughput and material costs. This value will not be clearly discernable in the monthly accounting reports. Odds are the energy costs across the other 39 process units will vary during the month, completely obscuring the actual gains.

The net result is that although an ROI projection is required to get project funding, the ROI is seldom determined after the project is executed. Initial projected ROIs for automation projects set an expectation for management as to what benefits can be expected. The lack of post-project ROI determination dashes those expectations. This contributes to management's negative perception of the value of automation.

Making automation project value visible

The solution to realizing the true potential value from brownfield automation projects is to overcome each of the three barriers presented above, but in reverse order. Once brownfield automation projects are shown to be valuable, the general value from automation will become greater and more discernable. When management starts seeing significant returns on automation project investments, the entire attitude toward automation project value will change for the better.

Overcoming the third barrier, lack of appropriate business measures, can be accomplished by implementing real-time accounting measures right down to the process unit level in industrial plants. Interestingly, real-time accounting

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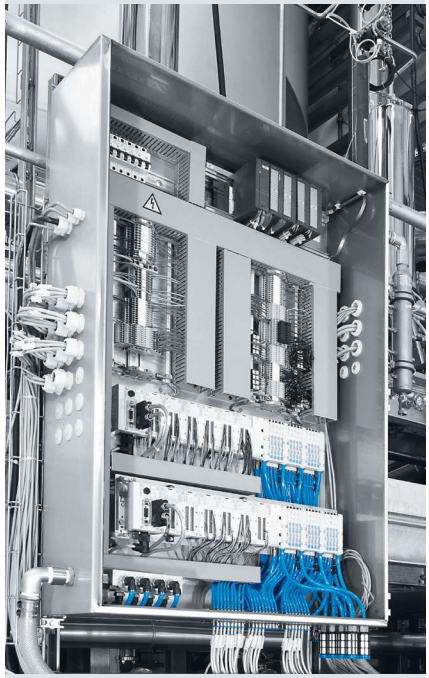

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measures are calculated using the plant sensor-based data in conjunction with plant business data. This can be accomplished in the control systems that have real-time access to the sensor-based data. Extending control systems to incorporate real-time accounting measures is a critical first step. Once real-time accounting algorithms are operating in the control systems, they can be historized into standard process historians to produce a real-time accounting system. These real-time accounting measures can be used to discern the value of any activity right down to the process unit level in real time. As an automation project comes on line, the real-time accounting system will measure the financial impact of that project. Making these measures visible to management is a first step in realizing the automation project value potential.

Once management can see the potential value from automation, it may be more inclined to adjust existing capital budget and project management processes to include an incremental business value-add metric along with the traditional on-time, on-budget project team performance measures. The other project management changes described above need to come along with this shift in project team performance measures.

Once management can see the potential value from automation, it may be more inclined to adjust existing capital budget and project management processes to include an incremental business value-add metric along with the traditional on-time, on-budget project team performance measures.

But with these changes in place, industrial companies will unleash one of their most powerful performance-impacting resources, project teams, on the problem of providing incremental business value. Project teams empowered to produce this value will undoubtedly succeed as they have in other areas.

With real-time performance measures in place and project teams empowered (and measured accordingly) to add business value, replacement automation will naturally be supplanted by business value-adding automation. Complete financial visibility across the automation project cycle, as compared with mere cost visibility, will give operations teams, project teams, and, most importantly, managers a huge change in perspective with respect to the great business improvement potential from automation projects.

Automation is one of the greatest potential value-affecting capabilities available to industrial companies. It is time to change our approach to performance measurement and automation projects to finally drive the value from automation that has been expected and available for decades. ■

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HMI design

Baker's dozen tips to create more effective HMIs

Figure 1.
Talking to operators and observing their interactions with screens are essential steps when designing HMIs.

By Chip McDaniel

There are many ways to develop human-machine interface (HMI) screens for machine and process automation applications. Effective implementation requires discipline in design. The look, feel, and ease of use of an HMI can vary widely given all the tools, object libraries, animations, and colors available with modern HMI software—but there are guidelines to improve HMI effectiveness.

Guidelines, standards, and handbooks covering HMI design include those published by ISA, ASM, ISO, and NUREG. All these resources discuss a wide range of design, build, operation, and maintenance methods for effective HMIs. Many also discuss safety, quality, reliability, and efficient control of the equipment or process under normal and abnormal conditions.

These standards can be a basis to create internal company HMI design guidelines, which in turn can be used to create consistent and effective HMI screens from one machine or process to the next. Standards are important, and users should consult them before final implementation of an HMI design, but they can be difficult to read and interpret. The tips below are provided as design aids.

Create a storyboard

A good starting point for HMI design is a text-based outline documenting each screen's proposed content. With the operator's point of view and ease of use in mind, the designer creates storyboards to flesh out the details of the main screen, equipment status screens, set point or

recipe screens, manual functions, message displays, fault displays, and other ancillary screens. The designer can then convert these text outlines to a storyboard for each screen. The storyboards should highlight dynamic graphics, such as status indicators, and should include repeated graphics such as titles at the top of the screen, along with navigation buttons at the edges of the screen. Align and group screen buttons, indicators, and numeric displays as appropriate.

Talk to operators

While creating the storyboards, ask yourself what is important to the operators, or better yet, ask the operators if they are available for consultation. Having more data on the screen is not always an improvement, so do not overwhelm the operators with information. Instead, focus on their tasks and what is needed to understand the state of the machine or process. You do not need to display every analog value, but screens should instead present information so that overall operating status can be understood at a glance.

Have an experienced operator review the proposed storyboards. Many machines and process skids are built by original equipment manufacturers (OEMs) and sold to customers. If this applies to your situation, now is the time to engage your customers and their operators in the HMI design to get their input and incorporate their specific requirements.

Deploy a prototype of the HMI as early as practical in the design cycle, and watch how the operator uses the interface. If it is not possible to develop and test a prototype, use the storyboards for a mock-operation session with the operators.

Look for awkward situations where an operator may be struggling to interpret what he or she is seeing, and look at the number of actions necessary to carry out a common task. Are there additional features you can add to relieve operators of needless button presses or otherwise make them more efficient?

Ask the operators for their advice and opinions, but filter their comments. They may not always be tuned into the big picture. At the same time, realize that your detailed knowledge of the machine or process may cloud your perspective of what is actually required to operate the machine efficiently. Never forget that you are building the HMI for their use, so their input is very valuable (figure 1).

Use color and animation judiciously

Many HMI guidelines recommend limiting the use of color and using low-contrast gray backgrounds to create less cluttered screens. For example, a light gray screen background where a typical indicator would be dark gray in the *off* state and white in the *on* state is easy on the eyes, and makes sense intuitively, as a light bulb turns white when on. In general, use subdued colors or shades of gray, and reserve bright, saturated colors to indicate abnormal conditions.

FAST FORWARD

- There are many standards for the design of machine and process automation HMIs, but interpreting and applying these often opaque documents is difficult.
- While not the last word, the tips offered here can help jump-start the HMI design process.
- These tips put an HMI design on the right path, but check final implementation for compliance with standards and customer requirements.

Be cautious when developing graphics, as this is not the opportunity to express your inner Picasso. Some more artistically talented HMI designers enjoy using all the colors and graphic animations available in the software, but this should be avoided. Showing pumps and fans spinning, valves opening and closing, and fluid moving in the pipes is cool—but may be excessive and can distract the operator. Only animate if it makes the operator more efficient.

For example, using animation to show the position of a part on a transfer line might be an efficient and quick indication of production status. However, showing a pump motor rotating is a distraction if the intent of the screen is to show only high-level fault indications.

Add graphics

"A picture is worth a thousand words" could not be truer for HMI design. Operators really do grasp the meaning of an image faster than they can read text or interpret status from a group of colored buttons. And if there is a language barrier, pictures become even more important (figure 2).

Many HMIs allow overlapping objects, so consider allowing a small graphic to overlap a button. Excessively large graphics and animation effects can slow the performance of a particular screen, so limit their use or isolate them on alternate screens to ensure fast response on the primary screens.

Data display is a key component of HMI design, and different types of data require different display types. A number on a screen may accurately display speed, but the engineering units may be in question and the acceptable range unknown. Solve this by adding the units, inches per second for example, and the maximum and minimum values in a table. A line

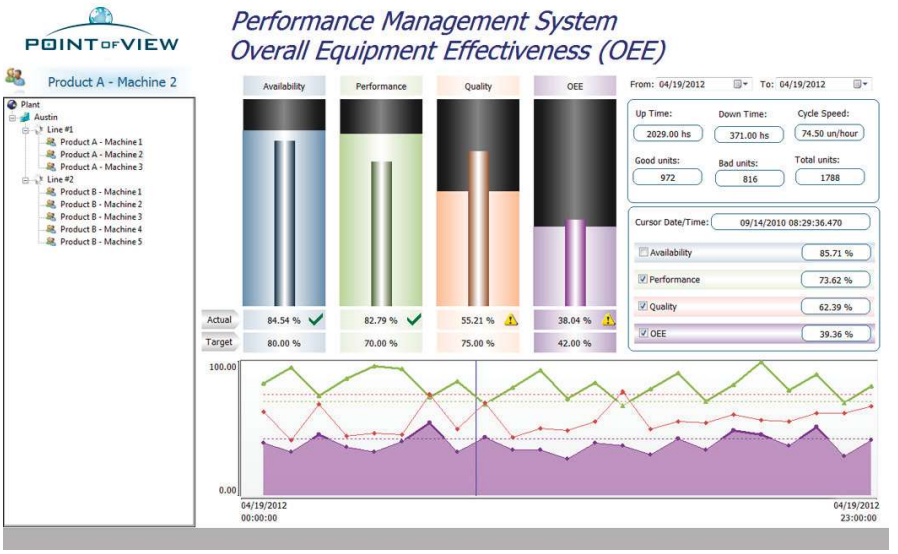


Figure 2. This HMI screen uses graphics well to display information in a format that operators can comprehend quickly.

graph with a trending function can display past and present data, and is a good indicator of future values, because an operator can quickly see values moving toward an upper or lower limit.

Display appropriate images

An image of the machine or process skid being controlled or other real-world images are incredibly useful and can help operators understand the location of problems such as blocked actuators or faulty sensors.

Consider how a modern office copier can lead a complete novice to the location of a paper jam in seconds by using clear images of the machine, location pointers, and arrows to the flaps and hatches that must be accessed to clear a jam. But resist the temptation to just import design drawings and diagrams, as this often creates a busy screen with excess detail (figure 3).

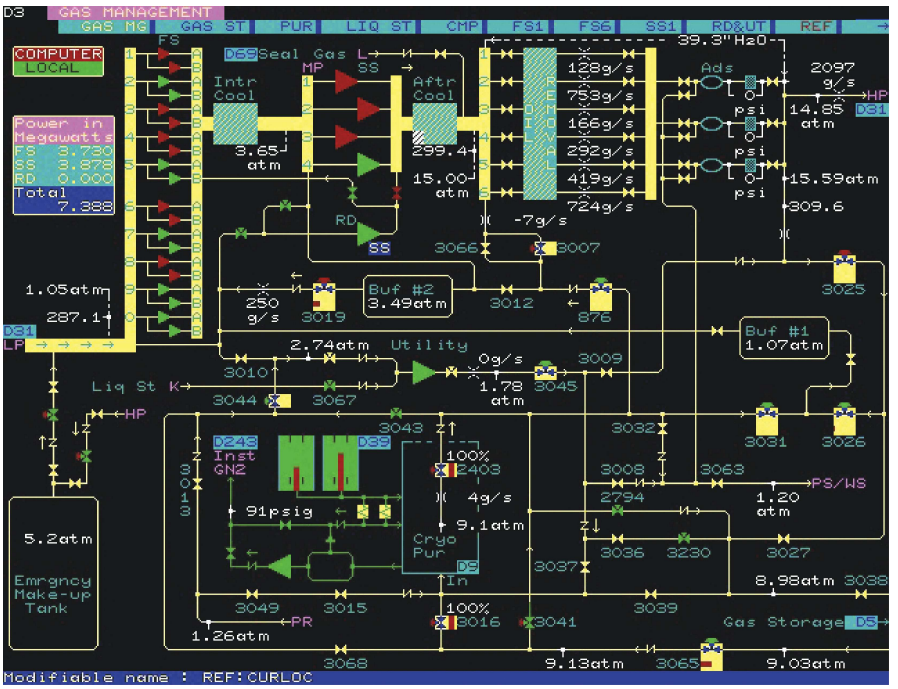


Figure 3. Although this screen contains lots of useful information, it is likely to cause confusion for the operator instead of creating clarity.

Keep important items available

Reserve a portion of the screen for important items such as stop, start, and key set points—so operators immediately know where to look for this critical information. Perhaps place a band at the top or on the side for these controls, and make sure this area is completely consistent on each screen where it is displayed.

Many HMI software packages offer a "background screen" feature that allows designers to create, edit, and maintain such an area in a single location within the project layout and then display it across multiple screens as desired.

Offer situational awareness

Make sure relevant data is displayed clearly so operators can grasp the current state of the machine or process at a glance. A good display answers two key questions "What state is the machine or process in now?" and "How does that compare to optimal conditions?"

For simple machines, displaying data with respect to the present state of the machine may be enough, but for more complex machines or processes, the screen design and layout should strive to help an operator foresee potential problems.

It might be enough to know the current temperature of the oven, but could a trend graph allow the opera-

HMI design tips

- Create a storyboard
- Talk to operators
- Use color and animation judiciously
- Add graphics
- Display appropriate images
- Keep important items available
- Offer situational awareness
- Limit required screen access clicks
- Give feedback and lead the way
- Beware of pop-ups
- Use date and time stamp logging
- Use appropriate password protection
- Create a style guide

tor to predict and prevent a problem? The answer may be yes, but do not assume adding more information is always better. Many layout and design decisions detract from situational awareness, and data overload and graphics that draw focus to the wrong information are two of the most common mistakes.

Limit required screen access clicks

Keep all screens within two to three clicks of the home or main screen. Consider the workflow for an application. Ask which tasks are performed most often, and which are seldom used. With this knowledge, optimize the layout to be as efficient as possible.

Keep screen menus and screen change actions as consistent as possible throughout the application, and always leave a clear and consistent path backward, so even the least experienced operator does not get lost.

Give feedback and lead the way

Give the operator visual and audible clues as affirmative feedback when buttons are pressed or when certain steps have been performed. This builds confidence and satisfaction with the system, and optimizes operator interaction.

Use color and animation to lead operators step by step through complicated processes if necessary, but be as simple as possible. For example, long lists of operator tasks indicate opportunities for additional automation or other workflow-based solutions.

When engaging with operators to evaluate the effectiveness of displays and functionality, two good questions to ask again and again are "Where were you expecting to find that information?" and "What were you expecting to happen?" Use their answers and other feedback to improve subsequent screen designs.

Beware of pop-ups

Use pop-ups sparingly, if at all. Nothing is worse than a cascade of pop-up error messages, all of which have to be acknowledged individually, before the operator can get to the screen displaying the root cause where the condition can actually be resolved.

Other techniques that draw the operator's attention are simply changing the background color of an object or a section of the screen or making objects blink, although these should also be used sparingly.

Overusing blinking or flashing habituates the operator to the effect, and thereby diminishes their effectiveness as an alert. They may even distract the operator from a more important issue elsewhere on the screen.

Use date and time stamp logging

Good HMI designs require more than just properly designed screens; they must also intelligently use features such as alarm and event logging.

Do not just track alarms and events, but also note when they occurred. Many recurring issues may seem unrelated—but often correlate to shift changes, coffee breaks, or even the startup and shutdown of nearby equipment or other periodic plant operations.

Alarm confirmation requirements can show you how quickly operators react to certain conditions. If some reactions seem particularly slow, consider how refining the HMI layout and design could quicken response.

Use appropriate password protection

Carefully consider which screens or objects should be available to all personnel and which should be restricted to maintenance, engineering, or management staff. Consider safety and the complexity of functions as you make the evaluation. Realize that an overly restrictive, complex, or ever-changing password policy can hurt productivity—and may even cause sharing of passwords among unintended personnel.

Create a style guide

Create a set of common styles for consistency across all HMIs in your plant and across multiple facilities. If you are an OEM, the style guide should create consistency among all of your products, as this will help train your technicians.

Using indicators, graphics, trend objects, and other graphical elements consistently increases operator familiarity and improves comprehension. Look for HMI software packages offering both an object library and a screen library to store and share proven elements or entire screens, respectively. Libraries let you easily reuse common items for multiple projects.

There is a frustrated artist residing in many of us, but HMI screen design is not the place to express these urges. Instead, use these tips to develop HMI design guidelines, and use these guidelines to consistently create effective screens from one project to the next. ■

ABOUT THE AUTHOR

Chip McDaniel (cmcdaniel@Automationdirect.com) works in technical marketing for AutomationDirect and is a graduate of the Georgia Institute of Technology. He has worked in the industrial automation field for 30 years, designing, building, commissioning, and marketing automation systems and equipment.

View the online version at www.isa.org/intech/20150405.

RESOURCES

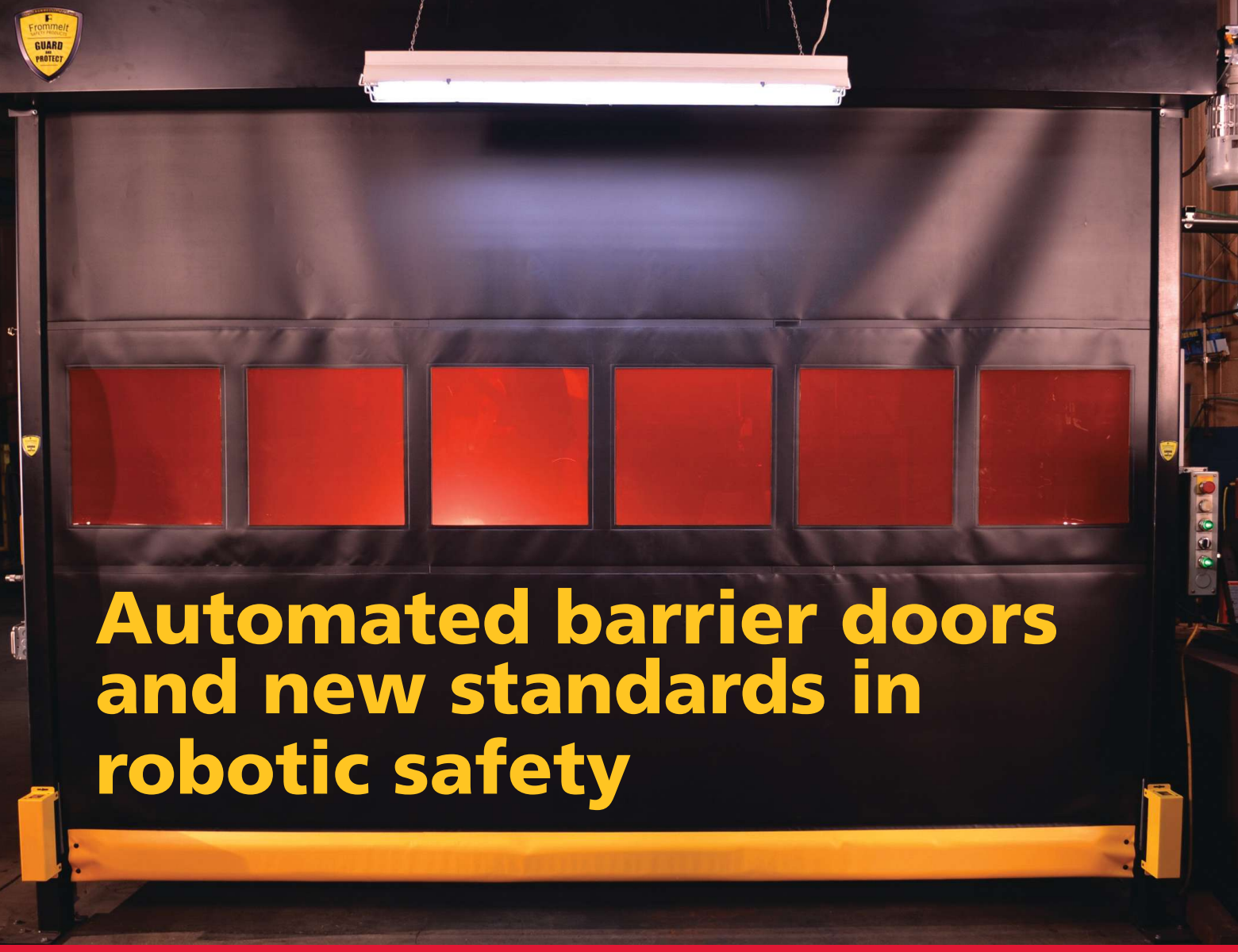
ISA Standards
www.isa.org/findstandards

"The high performance HMI"
www.isa.org/standards-publications/isa-publications/intech-magazine/2012/december/system-integration-the-high-performance-hmi

"HMI standards writers weigh in on importance"
www.isa.org/standards-and-publications/isa-publications/intech-magazine/2009/december/standards-hmi-standards-writers-weigh-in-on-importance

The High Performance HMI Handbook
www.isa.org/store/products/product-detail/?productId=116012

Designing for Situation Awareness: An Approach to User-Centered Design
www.crcpress.com/product/isbn/9781420063554



Automated barrier doors and new standards in robotic safety

Physical barriers stand up to latest OSHA and RIA safety standards for automated processes

By Eric Esson

The past few years have been marked by significant changes in industrial safety standards in the U.S. Safety managers, plant managers, and others who are not carefully monitoring these developments may not be keeping pace with the latest codes and regulations.

According to the Occupational Safety and Health Administration (OSHA), "machine guarding" that pertains to machines' general requirements and general industry (29 CFR 1910.212) consistently falls in the top ten most frequently cited OSHA standards violated in any given year. When combined with newly introduced safety regulations, it is easy to understand why this commonly misunderstood topic is more confusing than ever. The multitude of robotic applications and the growth of robot use and automation in all industries only worsens the problem.

Conducting a thorough risk assessment is the best way to maintain a safe work environment, especially when adding new automated processes. Thanks to the new Robotic Industries Association (RIA) R15.06-2013 standard, proper risk assessments are no longer just a best practice; they are mandatory.

RIA 15.06-2013 harmonizes international and U.S. standards

Adopted in May 2013, this new standard references ISO 10218-1 & 2, which addresses robots, robot systems, and integration. The new RIA 15.06-2013 was written to be compliant with international standards already in place in Europe, making life easier for manufacturers and end

users. This new standard requires better hazard identification related not only to robotic motion, but also to the task being performed. Additionally, it requires validation and verification of the safety systems employed and requires designs that incorporate protective measures for the robot cell and the operator.

Some of the biggest changes in the new RIA 15.06 industrial robot standard have to do with safety-rated motion and allowing advanced programmable safety devices to be used. What this means is software will now be allowed "safety-rated" control of various aspects of the robot's function, limiting the area in which the robot operates and the speed of robot motion. This is a departure from older standards in which programmable safety controls were not allowed.

In addition, as part of this standard, risk assessments are now required. Many professionals responsible for plant safety have been conducting risk assessments to increase safety as a matter of practice. These new regulations mandate risk assessments be conducted.

The basics of machine guarding risk assessment

Understanding and assessing these risks—and ensuring compliance—is not a simple task. The first step for facility/safety professionals is to identify and understand all applicable codes and regulations for their facility and operation. Next, they should examine the prevailing machine guarding choices for those applications to validate their safety system and its components. Although many guarding methods and products are available, not all can be applied universally.

Every machine guarding application has a set of unique challenges and associated risk. The choices a facility manager makes for one application might not be the same, or appropriate, for the next. In most cases, safety-conscious managers would not guard an industrial robot the same way they would guard other equipment, because the risk associated with each differs greatly. Risk may even vary between similar operations, depending upon employee exposure and other factors.

Determining risk

When performing a proper risk assessment, point-of-operation guarding is the most involved aspect. It is easy to place perimeter guarding around the entire process. However, in most situations a machine operator needs to interact with the process by loading or unloading materials (such as metals to be welded) and "running" the machine.

This point-of-operation is where things get tricky. Many details must be considered when it comes to this area, including the layout or design of the process and the limits of the system. Also, facilities must properly identify all associated hazards and devise methods for hazard elimination and risk reduction.

FAST FORWARD

- Risk assessments are mandatory. Are you ready?
- Safety standards are becoming more quantitative.
- New standards support added safety and increased efficiency.

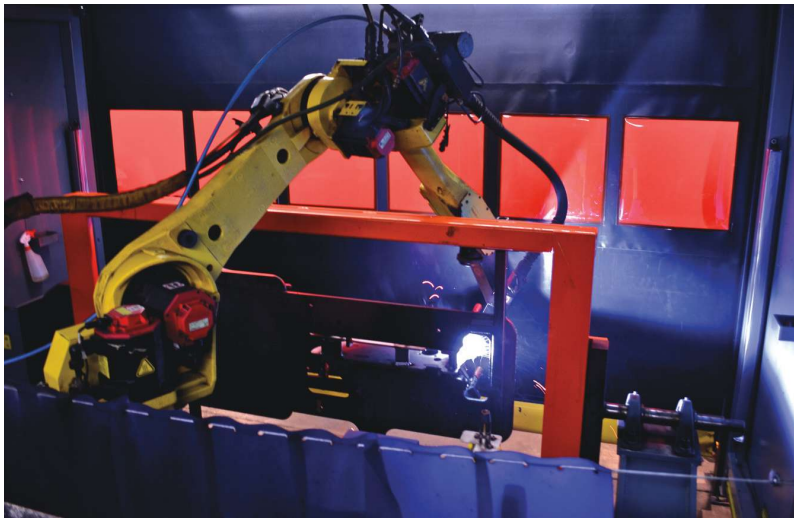
Once the severity of the potential hazard has been determined, the frequency or duration of exposure and the possibility of eliminating or limiting exposure can help safety managers choose the proper machine guarding device. Also, using the distance formula identified in OSHA guidelines can help in this selection. Per this formula, the safeguarding device has a prescribed location based on a number of factors, including secondary hazards that might harm a machine operator.

Limiting hazard exposure

Light curtains, laser scanners, and other presence-sensing devices are a commonly used and widely accepted method of machine guarding in manufacturing facilities from Tier 1 automotive to small machine shops and fabrication facilities. With presence-sensing, the automated process ceases once the safety device's infrared beam is tripped.

In many instances these devices provide acceptable safety. However, they are not always the best choice in all applications, especially after a risk assessment is performed.

Curtains may be the right choice in some



A fast-acting automated barrier door or roll-up curtain eliminates exposure to dangerous movement machines and hazards produced by the process, such as smoke, flash, splash, mist, and flying debris.

applications. However, fast-acting automated barrier doors or roll-up curtains may be better choices because they can eliminate exposure to both the dangerous movement of the machine and the secondary hazards produced by the process, such as smoke, flash, splash, mist, and flying debris. This further diminishes the potential risk and the severity of exposure.

Coupled with safety interlocks (up to PLe per EN ISO 13849-1 when integrated properly), automated barrier doors and roll-up curtains offer an increased level of protection for point-of-operation guarding. They restrict access to the process and contain secondary hazards of automated welding operations by placing a barrier between machine operators and machine movement.

These types of guards are an ideal alternative to light curtains and other presence-sensing devices in many situations.

2012: From EN 954-1 to ISO 13849-1 and EN 62061

One of the biggest regulatory paradigm shifts in recent memory occurred in 2012 with the move from EN 954-1 to ISO 13849-1 and EN 62061. Support for this change began in 2007, but its opponents were able to put the change on hold in 2009 and kept it there through 2011. Although approval of this harmonized standard was a hotly contested fight, it is now here to stay. Fortunately for those in charge of safety, best practices and market-ready solutions already exist.

ISO 13849-1, when broken down to the basics, provides a clearly defined set of rules to follow when designing the safety system as applied to industrial machine control systems. Officially defined as "safety of machinery, safety-related parts of control systems, general principles for design," this regulatory shift was made necessary by increasingly complex manufacturing processes using robotic and automated technology. Safety control systems and methodologies were forced to keep pace.

The ISO 13849-1 standard is more quantitative than EN 954-1. It applies common sense and forces facility managers to validate their safety systems, whereas EN 954-1 was conceptual and only required facilities to apply safety devices (controls) properly, specifying nonprogrammable, out-of-date technology.

Let's face it, our increasingly complex manufacturing processes require more complex systems to monitor their safe operation and keep machine operators safe. Automated processes, robotics, and even time-tested processes all require considerable attention to ensure those processes can proceed both efficiently and safely. EN ISO 13849-1 is ultimately making a much safer manufacturing environment, because it accounts for the reg-

ulatory gaps in the older standards.

Because every robotic system is different and has its own set of guidelines, it is important to realize what they are before implementation. Know specifications such as space and cycles. Integrators of new robotic systems will be required to perform these risk assessments in an attempt to identify potential dangers and ways to limit and eliminate them.

RIA 15.06 is similar to ISO 13849-1 in that it has a quantitative approach to hazard identification. A functional safety requirement of D (performance level "d") will be required of all robotic systems, as well as structure category 3 (dual channel), unless a risk assessment determines otherwise. PL safety and category ratings will offer a much more measurably reliant way to gauge safety.

The new standards in safety

As new regulations like RIA 15.06 and EN ISO 13849-1 are adopted, it is important to keep up with the latest and greatest safety technologies available to match the right product to the right process. Consider not only potential machine hazards, but also the task being performed. Advances in design and available technology make automated barrier doors ideal for guarding the machine and protecting operators, ultimately increasing productivity and the level of safety for years to come.

Creating a smaller manufacturing cell

Due to the nature of a properly interlocked automated barrier door, certain aspects of OSHA's safety distance formula become moot, because there is no depth penetration factor. The safeguard can be placed much closer to the hazardous area, and there is less space dedicated to a "safety zone," which reduces the manufacturing cell's footprint. This space savings is a huge benefit in most facilities. The smaller safety zone may also help to increase productivity and create a better ergonomic situation for the machine operator by limiting required motion.

Eliminating accidental entry into the cell is another benefit of interlocked automated barrier doors. Because their safeguarding can be seen (unlike the invisible infrared beams of presence-sensing devices), they greatly reduce the opportunity for accidental work stoppage. The physical separation is a clear visual indicator that the machine operator needs to be on task.

Do not fall out of compliance in 2015

Regardless of the safety device selected for machine guarding, facility managers must remember to perform a proper risk assessment. Although it can be tricky, the process will ultimately make a facility safer for workers and keep it in compliance with RIA R15.06, which is now in effect. ■

ABOUT THE AUTHOR

Eric Esson (eesson@ritehite.com) is the national sales and marketing manager at Rite-Hite Machine Guarding, which provides safety equipment for welding, robotics, and other industrial processes.

View the online version at www.isa.org/intech/20150406.

When is remote management the right move?

By Paul J. Galeski, P.E., CAP

ABOUT THE AUTHOR

Paul J. Galeski, P.E., CAP (paul.galeski@mavtechglobal.com), the CEO and founder of MAVERICK Technologies, specializes in high-level operational consulting and the development of automation strategy and implementation for automation technology. He is also involved in expert witness testimony and is a contributing author to Aspatore Books' *Inside the Minds*, a series of publications that examines C-level business intelligence.

Remote monitoring technology has grown in sophistication by leaps and bounds over the past few years due to the proliferation of high-speed Internet and smart mobile devices. Company engineers and executives can now look securely at a plant through its automation systems and see exactly what is happening in real time from anywhere in the world. Although remote monitoring has progressed, many plants and companies could benefit from the next step, to active and systematic remote management by a skilled automation solutions provider.

Remote management adds critical elements to improve upon simple remote monitoring, allowing appropriately authorized individuals to be a virtual extension of your control room, anywhere in the world. These people can monitor your automation systems, help diagnose problems, tune loops, optimize processes, and generally improve production. They could theoretically be your own employees located remotely, but the opportunities for hiring another company to assist with your plant is growing in practicality and popularity.

Some number of automation professionals with a deep understanding of your processes and how they work should be on your payroll, but individuals with similar and other skills can be outsourced. The number of people needed in your plant at any given time may be far fewer than you think if you can supplement their activities with remote management personnel. The original equipment manufacturer (OEM) of your main automation platform may be able to assist. However, most plants are a mix of systems from a variety of suppliers, which can cause issues for the OEM when it comes to holistically solving operational problems.

An automation solutions provider without ties to a single OEM can oversee the functioning of these mixed platforms and ensure they work together seamlessly. Moreover, as companies strive to maintain the lowest possible headcount, being able to extend an internal staff with remote managers and operators can fill critical gaps when needed. Plant operators and engineers can then concentrate on their assigned tasks, while other aspects of automation system maintenance and plant operation are managed remotely.

What can you expect from remote process managers? Probably more than you realize. For example, they can perform the following tasks:

- Identify and diagnose problems before they cause production interruptions
- Help get production going again more quickly after a failure
- Monitor networks
- Check system health routinely
- Provide oversight of routine system backups so operational improvements are captured and preserved
- Keep cybersecurity protection up to date by supervising software updates and patches

When first considering the advantages of remote management, some companies only consider the cost. Any expenses must be offset with savings from reducing the need for internal staff to perform critical tasks, however. Another factor is the cost of lost production, unsellable product, or safety lapses, much of which can be saved by active remote management.

Your internal people can probably solve most problems given enough time, but what is production in your plant worth? How many problems happen when you are the least prepared to deal with them? Perhaps your staff could have the plant running again in five or six hours, but how much would it be worth to have it running again in one hour, or better yet, to avoid the problem entirely? Your company may find the production gained by having highly qualified help available 24/7 vastly outweighs the cost.

Working with a lean staff puts many plants at risk. In many cases there simply are not enough qualified and experienced people, and new hires may not have the skills necessary to maintain desired production levels. Remote management can fill the gap by delivering boots on the ground quickly as the situation demands it.

Training costs for remote management staff are minimized, because it is the service provider's responsibility to get its staff up to speed on a plant's specific processes and operations. Remote management personnel have to become familiar with the unique aspects of a client company's facilities, down to specific systems and pieces of equipment. Individuals from both sides develop a level of comfort as they get to know their opposite numbers.

If this kind of solution is what you are looking for, there are many ways to explore your options, and it is possible to start small and evaluate before making significant investments. If the benefits are as expected, the remote management solution can be expanded to encompass more processes, plant areas, and automation systems. ■

Wisconsin cheese processor improves SQF standards with paperless initiative

By Heather Simcakoski

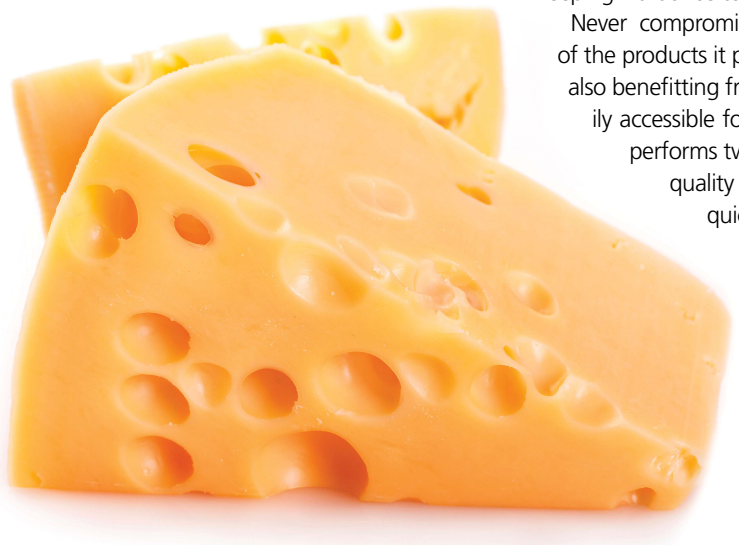
Motivated by new federal government regulations for food safety and inspired by their own commitment to safe quality food (SQF) standards, the twelfth largest dairy processor in the world engaged the services of a certified member of the Control System Integrators Association to implement a completely computerized data collection, storage, and reporting system. This dairy processor is based in Wisconsin, known as "America's Dairyland," but does business in all 50 states of the U.S. and in more than 60 countries worldwide.

The dairy began paperless reporting at one Wisconsin manufacturing plant—with immediate plans to implement the initiative throughout the company. The paperless system began saving time and expenses right away, reducing data entry errors, increasing efficiency, and reporting more accurate information. In one process alone, the processor initially reduced manual data entry by 80 percent. The savings added up quickly when multiplied over several plants and the dozens of processes at each that were previously performed manually.

The solution includes accurate and timely reporting by incorporating process data already gathered from an existing control system's data collection processor and deploying a new document retention solution, along with custom reporting. The scalability of the records solution allows collection and reporting on a wide array of process data. Additionally, the system's flexibility enables data collection at various points in the operation, including in the lab, at cookers, directly at each cheese vat, and during packaging.

Today, the dairy can follow the data pertaining to any batch of cheese through the entire records process—including the four-to-five data entry points where information was previously collected on paper by hand. Data is no longer manually recorded at each step in the process throughout the plant. Record keeping has improved. It is no longer subject to time delays, errors, and information redundancy. All information for a production run is now captured electronically, time stamped, and archived, allowing accurate and immediate retrieval and analysis. Production records are also retained for ongoing government-required record keeping—a bonus to SQF procedures.

Never compromising on the quality of the products it produces, the dairy is also benefitting from having data easily accessible for the mock recalls it performs twice a year as part of quality control. Now it can quickly and accurately pinpoint lot, vat, and ingredient data, to identify and isolate product issues. This will benefit the dairy's future production.



In addition to gaining daily operations efficiency and audit procedure improvements at the local plant, electronic data capture also benefits the corporate office. Plant data is now sent electronically, rather than by the old system of a handwritten submission recorded in Excel and later reentered into an enterprise resource planning (ERP) system. Custom reports generate only the information the operator needs for ERP entry. This one process alone saves the dairy several hours of manual data entry each week. This change improves data availability, reduces clerical errors, and saves hundreds of labor hours each year.

All information for a production run is now captured electronically, time stamped, and archived, allowing accurate and immediate retrieval and analysis.

From a single station to a production line to implementation across the entire enterprise, the cheese producer is now able to meet and exceed its SQF goals thanks to powerful data collection, reporting, and trending tools. It has gained critical insight into the performance parameters that help it improve yields, enhance quality, reduce expense, and ultimately grow profits—offering production within an efficient, paperless environment. ■

ABOUT THE AUTHOR

Heather Simcakoski is CRM analyst at Engineering Solutions Experts, Inc. The company specializes in customized, individually engineered automation of manufacturing and processing plants, including solutions for food and beverage applications. It is a certified member of the Control System Integrators Association (www.csiaexchange.com).

Nominations for Celebrating Excellence awards program open

ISA's Celebrating Excellence awards program showcases the remarkable technical accomplishments of individuals and companies in the automation community and acknowledges their dedicated service.

ISA has a long and proud tradition of honoring those who have contributed significantly to the advancement of the automation profession. Help us keep this tradition alive! Nominate at least one individual or company—among both ISA members and nonmembers—whose accomplishments in leadership, technical innovation, and education have moved the automation profession forward.

The nomination period continues through 30 April 2015. To nominate the peer, colleague, or company that you believe deserves a salute, log in to www.isa.org, select the "My ISA Account" button in the upper right corner, choose "Recognitions and Awards," and select "Submit Nominations" from the right menu. Celebrate the outstanding contributions and excellence of those in the automation profession. Nominate a peer, colleague, or company today! ■

In memoriam

Louis George Bourgois, Jr.

Louis George Bourgois, Jr., of Paducah, Ky., died 15 February 2015. In 1945, Bourgois was inducted into the U.S. Army Signal Corps and stationed at Walter Reed Medical Center in Washington, D.C. He was honorably discharged in April 1946 at the rank of private first class. After his discharge, he attended the Central Radio and Television School in Kansas City, Mo., and later worked as a machinist apprentice at Paducah's Illinois Central Railroad shops. In April 1952, he was employed by Carbide and Carbon Chemicals Corporation (subsequently Union Carbide Corporation Nuclear Division) as an instrument mechanic apprentice and, in 1954, completed his journeyman training. He then worked for Martin Marietta Corporation for more than 34 years, retiring from the Paducah Gaseous Diffusion Plant in November 1986.

Since 1957, Bourgois was a member of the Paducah section of ISA and served as its treasurer 1962–2008. In 1974 he began serving as ISA Paducah section education chairman. He was influential in establishing an ISA scholarship program for engineering students entering Paducah Community College and an ISA student section with its faculty advisor Mr. George Washburn. In 2000 he helped facilitate the development of an instrumentation technology associate degree program at West Kentucky Community and Technical College and coordinated manufacturers' equipment donations and construction of training resources by Paducah section members.

In recognition of his service to the instrumentation technology profession and to higher education, ISA awarded him the Distinguished Society Service Award (1993) and the Golden Achievement Award (2000). In 2002 he received ISA's highest honor for education, the Don W. Thompson Award, "in recognition of his efforts and achievement to encourage and support student activities and to mentor students." In 1994, he was recognized by the Kentucky State Board for Adult and Technical Education "for outstanding contribution and valuable service to Kentucky Vocational-Technical Education."

In memoriam

Michael John Yates

Astute, independent of mind, and quick of humor. Michael Yates, who passed away 23 February 2015, was all of these and more. Yates was secretary of the Institute of Measurement and Control from June 1987 to April 2008. Educated at Frensham Heights, an independent boarding school near Farnham, Surrey, U.K., and at Sussex University, where he read for his masters in history, Yates was erudite and articulate. This, with an unassuming manner and the ability to empathize, meant that his company was invariably valued.

Testimony to this was his worldwide network of contacts and colleagues from his many years' involvement with the international measurement community, particularly with ISA and the International Measurement Confederation (IMEKO). Glenn Harvey, former ISA executive director, shared, "I always had a special fondness for Michael. I think our respective positions as contemporaries resulted in a special bond. I always looked forward to, and looked fondly back on, the many meetings we shared at Gower Street, elsewhere in London, Brussels, Dusseldorf, and here in the U.S." Professor Tilo Pfeifer of Aachen University and IMEKO adds, "We have lost with Mike a likeable and highly experienced old friend and colleague. His lack will not be easy to fill, not only because of his professional skills, but mainly for his character and as a human being."

Yates joined the Institute in June 1973 and was appointed secretary fourteen years later. He was comfortable with the ethos of professional institutions and was acutely aware of the egos and sensitivities that can manifest themselves in such organizations. Not surprisingly, he had a strong commitment to protocol and constitutional affairs, but his overriding interest was in the Institute's learned society activities and the work of its technical panels. Given this and living in Teddington, London, he was a regular and welcome visitor to the National Physical Laboratory. Very much his own man, there were certain times when he drew the line and would not budge, whatever the issue. Our world is poorer for the passing of this likeable man.

ISA Certified Automation Professional (CAP) program

CAP question

A return on investment (ROI) ratio is used to evaluate the purchase price for an automation system and all other initial costs associated with the project against the accumulated cash inflows. **What is the primary pitfall in using this ROI ratio to make a decision as to whether or not to invest in an automation project?**

A. It fails to capture the qualitative benefits derived from automation systems.
B. It relies on vendor estimates of useful life.
C. It ignores equipment reliability and system maintainability costs.
D. It ignores the time value of money.

CAP answer

The correct answer is *D*, "It ignores the time value of money."

ROI can be quantified by:

$$ROI = 100 * \left[\sum_{i=1}^{n} (CF_i) / n \right] / (IC)$$

where,

IC = initial cost

CFi = operating cash flow, year *i*

n = project operating lifetime, years

In the above equation, qualitative benefits derived from the automation system (answer A) less maintenance costs (answer C) are included in the operating cash flow (*CFi*). The vendor's estimate of useful life (answer B) is taken into account by the project operating lifetime (*n*).

This leaves answer D, which is the time value of money. The ROI equation does not include a term to account for the lost opportunity to collect investment interest on the money that is being diverted to fund the automation project or the cost of borrowing the same amount of money from a lender (time value of money).

Reference: Trevathan, Vernon L., *A Guide to the Automation Body of Knowledge, Second Edition,* ISA, 2006.

Certified Automation Professionals (CAPs) are responsible for the direction, design, and deployment of systems and equipment for manufacturing and control systems.

ISA Certified Control Systems Technician (CCST) program

CCST question

You have a standard pneumatic instrument loop with a span of 200 units. **What is the steady state gain?**

A. 0.06 psig/unit
B. 0.07 psig/unit
C. 0.08 psig/unit
D. 0.12 psig/unit

CCST answer

The correct answer is *A*, 0.06 psig/unit.

Gain is defined in control theory as the change in input divided by the change in output. In this case, the gain can be calculated as follows:

Gain = (full range change in input) / (full range change in output)

= (15 psig – 3 psig, for a pneumatic loop) / 200 units of change

= (15 – 3) / 200

= 12 / 200

= 0.06 psig/unit

Reference: Goettsche, L. D. (Editor), *Maintenance of Instruments and Systems, Second Edition,* ISA, 2005.

Certified Control System Technicians (CCSTs) calibrate, document, troubleshoot, and repair/replace instrumentation for systems that measure and control level, temperature, pressure, flow, and other process variables.

Network design fundamentals for the connected world

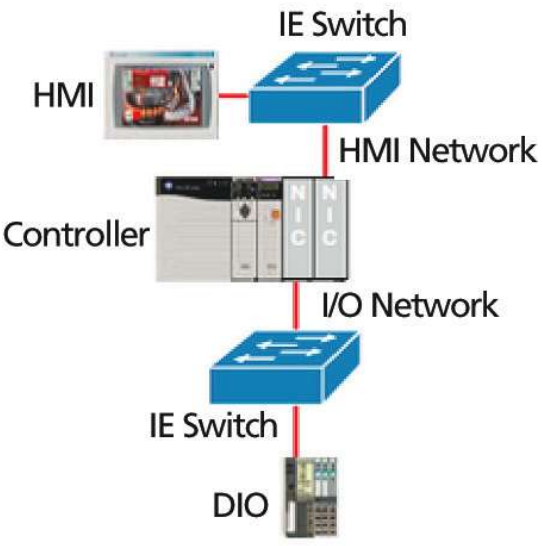
By Dan McGrath, P.E.

The benefits of converging industrial and information networks with a validated secure architecture based on standard Internet Protocol (IP) technologies are well established. The benefits include greater connectivity and integration across plants, easier data sharing across the enterprise, and better visibility into real-time operations.

Bringing together information technology (IT) and operations technology (OT) networks is complex because of the gray area between IT and OT roles and responsibilities within the company. IT and OT professionals must have a common understanding of a host of techniques and technologies to overcome this complexity and establish a converged infrastructure that is secure and manageable by all critical stakeholders.

The techniques and technologies used in network design can be simplified by leveraging the ISA/IEC-62443 Zones and Conduits Model developed by the ISA99 committee. The three design areas are:

- cell or area zone
- production site operations
- enterprise zone integration



Gateway physical segmentation example: two NICs for network segmentation

Designing for the cell or area zone

Several considerations must be made to ensure the network infrastructure addresses your data, security, and availability requirements at the cell or area network level. Machines and process skids are seeing high growth in number and criticality of IP connections. One of those considerations should be logical segmentation, which is the process of dividing end points into subnets and virtual local area networks (VLANs). A key recommendation for industrial networks is to create smaller layer 2 networks to improve performance with the maximum of 200 devices within a zone or VLAN.

Segmentation allows for smaller layer 2 domains, which helps constrain broadcast and multicast traffic. It also helps manage the network's real-time communication properties and supports the network's traffic-flow requirements. With segmentation, manufacturers and industrial operators can meet their security requirements by limiting remote expert or original equipment manufacturer network access to only specified machines.

Organizations often accomplish physical segmentation within the cell/area zone network by using separate cabling and switches. This common approach in Ethernet networks can become a hindrance to network performance if not properly planned. For example, physically separating input/output (I/O) and human-machine interface traffic without connecting the I/O traffic to an interconnected layer 3 switch can limit overall connectivity and even cause delays. Networks should, at a minimum, be connected to a layer 2 or layer 3 switch, rather than a controller, to interconnect.

VLANs are a very effective way to execute segmentation: specifically, for segmenting different traffic types—industrial and nonindustrial—as well as creating smaller layer 2 networks. Establish VLANs in a one-to-one relationship with subnets to make routing easier and more straightforward. Devices on a single VLAN are typically assigned IP addresses from the same subnet, and they do not require a layer 3 switch or router to communicate among each other within the VLAN. Using a layer 3 switch or managed switches with layer 2/3 functionality allows communication between VLANs. A management VLAN should be established for management across multiple cell/area zones.

Additionally, using structured cabling for interswitch links and more critical runs in your cell/area zone is a best practice network design approach. Point-to-point cabling is the norm

for connecting end-point devices in close proximity. However, more critical connections can benefit from using industry standards designed to ensure a testable, scalable infrastructure. A structured cabling approach has better organization and testability with patching fields, permanent links, and patch cords that are validated as a high-performance system for rising data rates and high availability. Structured cabling built on TIA-1005 or ENXXX standards have the bandwidth and noise immunity for challenging cell or area zone deployments to ensure uptime.

Designing for production site operations

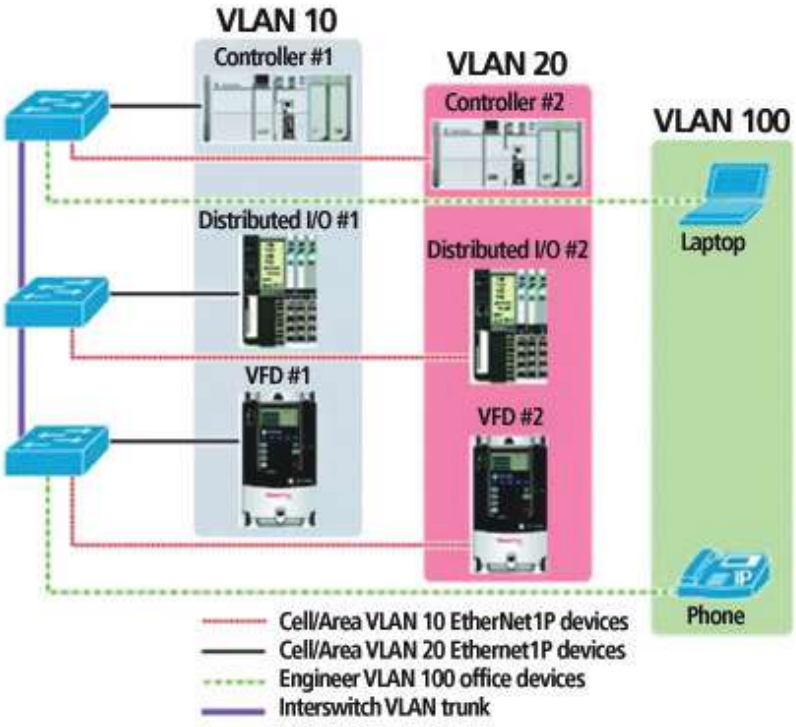
The Internet of Things (IoT) has created an explosion of smart IP–enabled devices that were not traditionally connected to the network. This has created an opportunity to deploy a more flexible architecture with mobile access to data and connectivity within the production environment. Wireless network technology is one of the key enablers for realizing the value of IoT. Wireless technology offers new capabilities, such as "bring your own device" and wireless security cameras, to protect assets and lower installation and operational costs.

Wireless local area networks are significantly different from wired LANs, and they should be designed for your security, reliability, bandwidth, and throughput requirements. For example, Wi-Fi Protected Access 2 with Advanced Encryption Standard encryption is the only mechanism recommended for control and automation wireless applications, and it should be used in combination with other security methods, such as preshared key authentication.

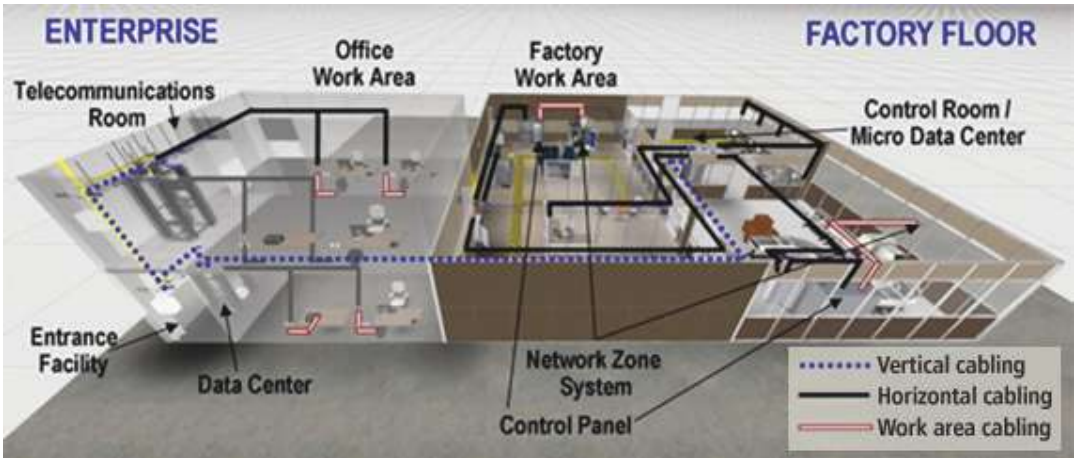
Wireless channel packet rates should be limited to 2,200 packets per second to help avoid packet delays, and they should be reduced in areas that experience interference and radio-frequency issues. Also, avoid the more heavily used 2.4-GHz band for industrial control applications because you may encounter interference. The 5-GHz band provides dedicated bandwidth and less interference. Conducting a site survey will also help you identify and curtail other potential interference within a production environment.

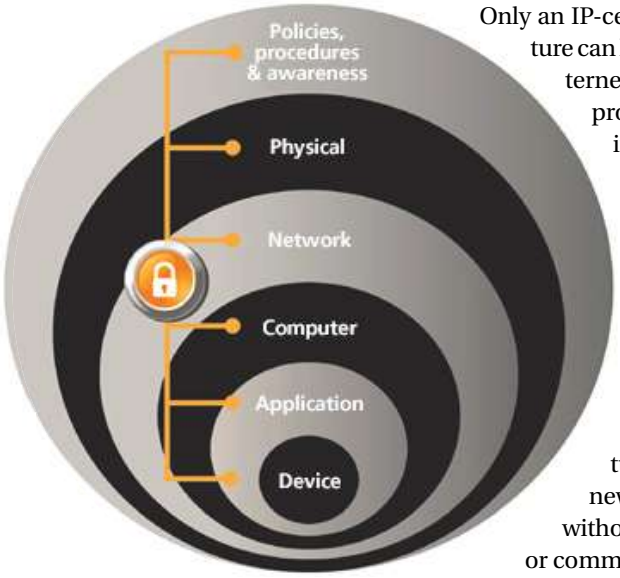
Virtualization is another key network technology in the connected industrial enterprise. Virtualization breaks the previously unbreakable link between hardware and software, so you can keep industrial applications running beyond the life cycle of their hardware. Virtualization also abandons the traditional one-to-one approach, meaning multiple applications and operating systems can run from a single server.

Important considerations when designing a virtual environment include your applications' RAM, CPU, and disk I/O requirements, as well as how many applications will be deployed in the virtual environment and the kind of network switching. Take into account your current



VLAN spanning multiple switches





needs, but also design for growth in the next five to 10 years.

You can engineer and build a virtualized system from scratch, but this approach can be costly and time consuming. You will need to purchase equipment from multiple vendors and spend weeks assembling, installing, and testing the system. An alternative approach is to purchase a bundled virtualization system with preassembled, tested, and validated hardware. Bundled systems also often come with support services, including on-site configuration and integration, which can shave days or even weeks off the deployment process compared to the build-from-scratch approach.

Keep in mind, however, that bundled virtualization systems are produced for a range of industries. It is important to select a system that is custom built for the demands of the industrial environment.

Enterprise integration

A truly converged IT and OT network architecture achieves seamless integration both horizontally, across multiple sites, and vertically, from the shop floor to the top floor. Doing this requires an understanding of the key techniques that bridge the gap between historically separate networks.

Using fully standard IP networks throughout the architecture, such as EtherNet/IP for automation communication, eases future application design. Only an IP-centric network infrastructure can help you better use the Internet of Things—which is the proliferation of connected industrial and commercial devices—because it is a unified digital communications fabric on which these IP-enabled devices can talk to each other. A fully standard IP network means your network design is more future proof and can deploy new services or applications without specialized gateways or communication stacks that create integration challenges.

Cloud computing also helps integrate the enterprise. Moving data centers to the cloud simplifies implementation and maintenance, reduces costs, and improves an organization's agility. However, the cloud still requires a robust physical infrastructure. Factoring in important design considerations—such as deciding what kind of cloud you will use and identifying your bandwidth and cabling requirements—will help you make the most of your cloud investment.

Enterprise integration also cannot occur without considering security. Given the expansion of the network fabric, increased access to sensitive data, and the breadth of cybersecurity threats, designing in multiple layers of protection should be considered a security best practice. This "defense-in-depth" approach builds a system of safeguards for both digital and physical security, putting multiple security measures in place to counter each individual threat.

Digital security measures include device authentication according to the 802.1X standard, policies that ensure the most recent antivirus updates and security patches are installed, monitoring key network statistics and log files for signs of intrusion, configuring network switches to manage traffic and access, and tightly controlling software used for remote access.

Physical security includes pass codes and access cards to limit personnel

access to rooms and connected machines. USB block outs help prevent unauthorized network access, the removal of sensitive information, and the threat of spreading a virus. Lock-in/block-out devices can prevent unwanted plugging and unplugging for copper and fiber connections.

Lastly, an industrial demilitarized zone can be a crucial barrier of protection between the enterprise and industrial zones. It serves as a choke point for all traffic between zones and can help better manage access, such as with authentication enforcement or by monitoring traffic for known threats. ■

ABOUT THE AUTHOR

Dan McGrath, P.E. (DTM@panduit.com), solutions manager for Panduit, helps lead Industrial IP Advantage, which is a coalition launched by Panduit, Cisco, and Rockwell Automation promoting standard, unmodified Ethernet and Internet protocols for industrial applications. McGrath has more than 25 years of experience in automation and industrial networking for global manufacturing operations. He holds a B.S. degree in electrical engineering and has attained professional engineer and ASQ certified quality engineer certifications. For more information and educational resources, visit www.industrial-ip.org.

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Preparing for the future: the human equation

By Doug Aldrich, Ph.D., CFM

Much has been written and forecasted about the various impacts of technologies on humans: work, pleasure, productivity, and behaviors. The exponential growth of technical tools and their explosive applications to our lives have been nothing short of astounding. These have created an equally dramatic increase in new jobs with required skills. Organizations seek to fill the new positions with internal staffing and training, along with outsourcing, to maintain a strategic edge.

As we prepare for our futures, we ask, "How do humans fit into the equation of coexistence with 'robots'?" (That question was reversed in past decades!) As technology grows in sophistication and complexity, it is vital to define new and improved skills, knowledge, and experiences (SKEs). Given unparalleled career doors opening, how do we view workforce development?

There simply are not enough people with the best skills, knowledge, and experiences to go around. These shortfalls cause problems with facilities, acquisition, retention, training/education expenses, and coaching/mentoring efforts.

Before we discuss that, let's touch on people supply. Most of us know about demographic "peaks and valleys" and the "less than two" replacement rates in Western countries. Shortages make it difficult to fill current positions, much less future, more demanding ones. This is crucial for mission-critical operations (MCOs) based on computers, automation, and networks. There simply are not enough people with the best SKEs to go around. These shortfalls cause problems with facilities, acquisition, retention, training/education expenses, and coaching/mentoring efforts.

Organizations should act upon the following:

- Determine *essential* jobs and resource them, *unimportant* jobs and eliminate them, and *bubble* jobs (either inside or outside) and evaluate them. Not all positions are strategic or valued, so difficult decisions have to be made. The end result is to focus resources on the SKEs valued for long-term needs. In most operations, except for manufacturing, people costs are about 80 to 85 percent of total expense budgets, with facilities most of the remainder.
- *Totally* change how they are supporting workforce development. It must be continuous, applied, seasoned, and subject to change. It is not about going to classes; it is about helping employees ensure education is relevant and useful. Assessing incoming SKEs is vital to understand gaps and putting together a "training" plan that makes sense to an individual and organization.
- Work to *integrate* functions so there are no silos. Information technology, human resources, finances, and facilities, for example, all have limits and trade-offs. This requires a new degree of information flow and cross-fertilization of roles; collaborative teaming will not be merely touted. Soft skills in communications and team management, although qualitative, need rigorous nurturing.

With those essential steps tackled, there are strong opinions about what courses will prepare people to work in the new age, especially in MCOs. Most begin with a science, technology, engineering, and mathematics (STEM) education at the A.A. or B.S. level. The nontraditional workforce suggests rethinking that and considering individual backgrounds. True training includes evaluation, usage, cultivation, and feedback for continuous improvement. New skills are added as the job moves forward; maybe antihacking training will become mandatory in the future.

An initial STEM hire will have some degree of depth. But here are my SKEs that need to be cultivated, if not initially present, for a sound future team member. Some are not surprises, while others may be:

- problem definition and solving
- imagination and innovation
- adept change management
- critical thinking and analysis
- multidimensional teamwork
- communications (all forms)
- process acumen

Although each of these can be defined, notice they may have little to do with chemistry, local-area networks, automation controls, or network design. That is because those technical skills may be only initial job requirements. It is the growth in them and those above that make the difference. Your organization will prosper with people "loaded" with my suggested SKEs.

"The factory of the future will have only two employees, a man and a dog. The man is there to feed the dog. The dog is there to keep the man from touching the equipment," said organizational consultant Warren Bennis. I used to be amused by this; now I am not so sure. We cannot abdicate the development of a competent and forward-performing workforce to a dog. ■

ABOUT THE AUTHOR

Doug Aldrich, Ph.D., CFM (doug.aldrich@comcast.net), is a Denver-based consultant on complex facilities with 40 years of industry experience and three decades of college recruiting and involvement.

HMI standard moves toward completion

ISA-101.01, *Human Machine Interfaces for Process Automation Systems*, is in the final stages of approval by the ISA101 standards development committee, Human-Machine Interface (HMI). The document was approved by ISA101 in a 2014 ballot, but with a number of review comments that have now been resolved. It is undergoing a final committee review before being submitted to the ISA Standards & Practices Board and then the American National Standards Institute. Final publication is expected by September of this year.

The purpose of the standard is to address the philosophy, design, implementation, operation, and maintenance of HMIs for process automation systems, including multiple work processes throughout the HMI life cycle. The standard defines the terminology and models to develop an HMI and the work processes recommended to effectively maintain the HMI throughout its life cycle. The target audiences are end users, designers, developers, and implementers of HMI systems.

The standard defines the terminology and models to develop an HMI and the work processes recommended to effectively maintain the HMI throughout its life cycle.

Use of the standard is intended to:

- provide guidance to design, build, operate, and maintain HMIs to achieve a safer, more effective, and more efficient process control system under all operating conditions
- improve users' abilities to detect, diagnose, and properly respond to abnormal situations

Looking ahead

With the nearing completion of the HMI standard, ISA101 will now focus on developing a series of ISA technical reports to provide further guidance in key areas of HMI, according to ISA101 co-chairs Maurice Wilkins of Yokogawa and Greg Lehmann of AECOM. This will be accomplished through working groups (WGs) tentatively planned as follows:

WG1 – HMI Philosophy, Style Guide & Design Guide

Provide the guiding principles and conceptual foundation for HMI design, including details on how the HMI is designed and used. The HMI style guide will provide HMI implementation details based on the HMI philosophy, feasible on all target platforms. It will use the guiding principles and concepts of the HMI philosophy to establish implementation examples and guidance.

The design guide will provide the foundation of the HMI system design, including the selection of the control platform and related operating system and choosing HMI tool kits to be used in the system.

WG2 – HMI Usability and Performance

Focus on performance factors and HMI usability. The performance factors can be viewed in two categories:

- HMI duty factors: factors related to display response, including such metrics as call-up times, display refresh times, and key clicks
- Performance-shaping factors: factors related to how fast an operator is able to detect, diagnose, and take action

WG3 – HMI for Mobile Platforms

Investigate the impact of mobile platforms. As mobile devices such as tablets and smart phones have come to the fore in recent years, questions have arisen about the applicability of the ISA-101 standard to these new platforms. This working group will investigate issues surrounding the applicability of the standard in such cases.

For more information about ISA101, contact Charley Robinson, ISA Standards, 1-919-990-9213 or crobinson@isa.org. ■

Intelligent device management committee update

The ISA108 standards development committee, Intelligent Device Management, emerged from a two-day planning session on 9–10 February in Orlando, Fla., with a new work plan for 2015 and beyond. The committee, co-chaired by Herman Storey of Herman Storey Consulting, LLC, and Kouji Demachi of Yokogawa, will develop standards and technical reports through a working group structure as follows:

- Configuration and Revision Management
- Diagnostics Utilization
- Procedure Management
- Calibration Management
- Intelligent Valve Management

Work descriptions for the working groups are in development, with that of the Configuration and Revision Management working group approved by ISA108 in early March "to document work processes for the management of intelligent device configuration integrity, configuration adequacy, and configuration congruency for a full facility life cycle."

ISA108 was established to define standard templates of best practices and work processes for implementation and use of diagnostic and other information provided by intelligent field devices. The committee is completing a ballot in March on its first work product, a technical report covering the concepts and terminology necessary to understand and communicate effectively about intelligent device management.

ISA108 has scheduled two additional face-to-face meetings in 2015:

- 27–29 April in conjunction with the ISA Analytical Division Symposium in Galveston, Texas
- 13–15 July in San Diego, Calif.

For more information about ISA108, contact Charley Robinson, ISA Standards, 1-919-990-9213 or crobinson@isa.org. ■

Focus on network design

Octopus OS30/34 switch

The Octopus OS30/34 switch has gigabit ports for connecting high-speed backbones and gigabit end devices with power-over-Ethernet (PoE) support. A bypass relay, which follows the IEC 61375 standard for Ethernet in trains, extends the redundancy mechanisms and broadens the functional range of the series.

Two customizable housings of the switch are available, allowing customers to choose a switch that meets specific needs. The mid-sized switch can have a maximum of 20 ports, including four gigabit ports and up to 15 PoE ports. The full-sized switch adds eight additional fast Ethernet ports for a total of 28 ports. The switch is available either with layer 3 software, including routing functionalities, or with layer 2 software for switching. Customers can choose fiber or copper cabling, and the switch has high-vibration resistance and broad protection to electrostatic discharges. With an IP67 rating, the switch meets the requirements of routing functions in waterproof and dust-tight housings for mounting outside of cabinets. It operates at temperatures ranging from –40°C to +70°C.

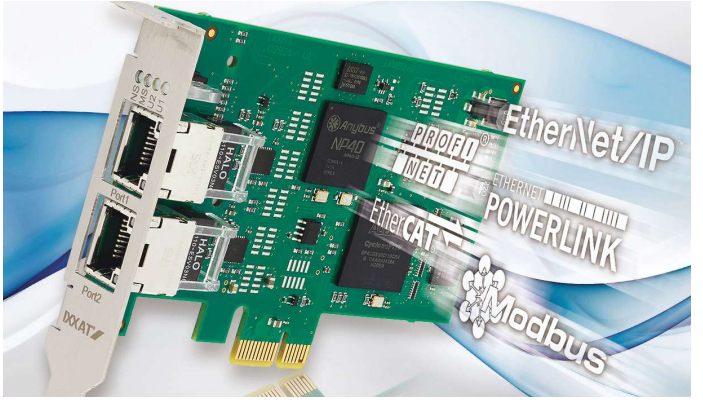
Belden, www.belden.com

Ethernet interface card

The IXXAT INpact Ethernet interface card lets PCs connect to EtherCAT, EtherNet/IP, Modbus TCP, PROFINET IRT/RT, Powerlink, and standard TCP/IP. The PCIe card supports multiple protocols for numerous industrial Ethernet standards and a uniform protocol and card-spanning application programming interface. Through the uniform driver concept, users can switch between protocols quickly and without extensive programming and also profit from future protocol extensions and developments.

The network solution inside the card is based on the Anybus CompactCom technology. Used in combination with a flexible FPGA-based PCIe bus connection, the Anybus NP40 network processor has all the functions required to process the communication between the industrial Ethernet network and the PC-based customer application. With low latency times and a deterministic real-time behavior, it is perfect for demanding industrial applications. The two-port Ethernet interface (10/100 Mbit, RJ45s) supports protocol-specific switch functions—such as IRT for PROFINET, DLR for EtherNet/IP, and Hub for Powerlink. With the Common Ethernet flash-update feature, the user can download any industrial Ethernet protocol without switching cards. The card is suitable for numerous applications, ranging from HMI systems, process visualization, and service tools to measurement, analysis, and test systems.

HMS Industrial Networks, www.hms-networks.com



VPN remote access routers

The Cosy 131 virtual private network (VPN) remote access routers have local-area network (LAN), Wi-Fi, and 3G+ connection options. Machine builders and original equipment manufacturers (OEMs) can access and troubleshoot machine programmable logic controllers (PLCs) and human-machine interfaces (HMIs) regardless of where they are in the world. The router has a USB port to connect to additional serial devices, an SD card reader for on-site configuration, and a configurable LAN/wide-area network (WAN) four-port switch. The device enables maintenance engineers to establish a secure VPN connection to the machine from anywhere via the company's cloud-based remote connectivity solution, Talk2M. The router connects on the machine network, alongside the machine PLC and the HMI, allowing remote connection from anywhere and from any laptop, tablet, or smartphone.

eWON, www.ewon.us

Industrial gigabit DIN-rail switches

The NT24k-8TX-POE and NT24k-16TX power over Ethernet Plus IEEE 802.3af/at switches have wire-speed performance in harsh environments. The hardened metal housing provides shock (200 g) and vibration (50 g) tolerance coupled with a –40° to 85°C operating temperature range. The NT24k-8TX-POE is an eight-port all-gigabit PoE+ switch with redundant 22–49 VDC power inputs for a built-in step-up power supply to boost power for applications such as security cameras, panel displays, and wireless access points. The NT24k-16TX is a 16-port all-gigabit switch with redundant wide 10–49 VDC power inputs for industrial applications. All NT24k models have N-Ring technology with 30 ms recovery for rings containing up to 250 nodes, automatic configuration, and backup and restore capability.

Red Lion, www.redlion.net

Ethernet switch

The EDS-405A-PTP Ethernet switch enables clock synchronization of automation systems to less than 1 microsecond. Time synchronization is critically important for decentralized systems in industrial automation, such as those found in chemical processing, printing presses, nuclear power plants, or any real-time automated process where multiple controls are distributed throughout the facility. The switch uses hardware-based time synchronization to fully comply with the Precision Time Protocol (PTP) specified in the new IEEE 1588v2 real-time clock standard. When configured as a boundary clock, the EDS-405A-PTP can act as a slave related to the grandmaster clock, while simultaneously working as a master clock related to its local slaves. The managed switch uses nanosecond-accurate time stamps to modify timing packets and synchronize clocks distributed over the network. The switch supports Modbus TCP, PROFINET, and EtherNet/IP protocols for SCADA/PLC management and monitoring.

Moxa, www.moxa.com

Industrial automation software

FactoryStudio Release 3 is a development platform built in the Microsoft .NET Framework, delivering software that runs on Windows PCs, Web browsers, tablets, virtual machines, and mobile devices. The software extends the concept of the real-time in-memory database used by SCADA and HMI factory-floor tools to a complete set of business entities able to deliver applications ranging from standalone HMIs to IT operations centers and management execution systems.

The release adds more than 5,000 additional vector symbols and flexible ways to link assets and tags with dynamic graphical representation. Get the variables and paste them into the drawing tool, and the display is ready to go. The software also has tools for importing tags and synchronizing with Rockwell PLCs, OSIsoft PI systems, PI AF, XML, SQL, or OPC servers. Communication protocols to numerous devices, annotations and alarms on trend charts, smooth filtering, recipe overlays, data exporting tools, Web gateways with Windows authentication or WS-Federation security, and tag-level access authorization are also included.

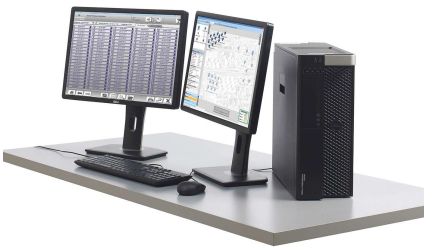
Tatsoft, www.tatsoft.com

Cellular router

The Ruggedcom RX1400 cellular router, a multiprotocol intelligent node, combines Ethernet switching, routing, and firewall functionality with various WAN connectivity options. The device is IP40 rated, does not use fans for cooling, operates continuously within a –40°C to +85°C temperature range, and comes with a metal housing that supports DIN rail, panel, or rack mounting. The router has a high level of immunity to electromagnetic interference, heavy electrical surges, extreme temperature, and humidity for reliable operation in harsh environments. The integrated GNSS (GPS/GLONASS) functionality allows the device to report its location as necessary for asset tracking.

Siemens, www.siemens.com

Smart wireless navigator software



Smart Wireless Navigator is a software platform to help users with large wireless deployments maximize their wireless networks. The navigator brings together smart wireless tools for planning, managing, and maintaining networks. Wireless network and device diagnostics and data are organized in an intuitive interface, along with the wireless tools. Users can manage their expanding wireless infrastructure and get the most value from their wireless networks. A single software platform design makes it easier for users with large deployments of wireless to manage their networks across functional groups, delivering actionable information to the people who need it. An intuitive design organizes large amounts of wireless diagnostic information and data, and existing infrastructure is illustrated and easily understood.

Emerson Process Management

Secure cloud service

Registration is open for the Secure Cloud Service. System integrators, engineers, and managers of industrial, embedded, and Internet of Things systems can do bidirectional supervisory control and integration, share data with multiple users, and have real-time access to selected data sets in a Web browser. The service can handle more 50,000 data changes per second per client, at speeds just a few milliseconds over Internet latency. The service allows industrial and embedded systems to securely network live data from any location.

Skkynet Cloud Systems, www.skkynet.com

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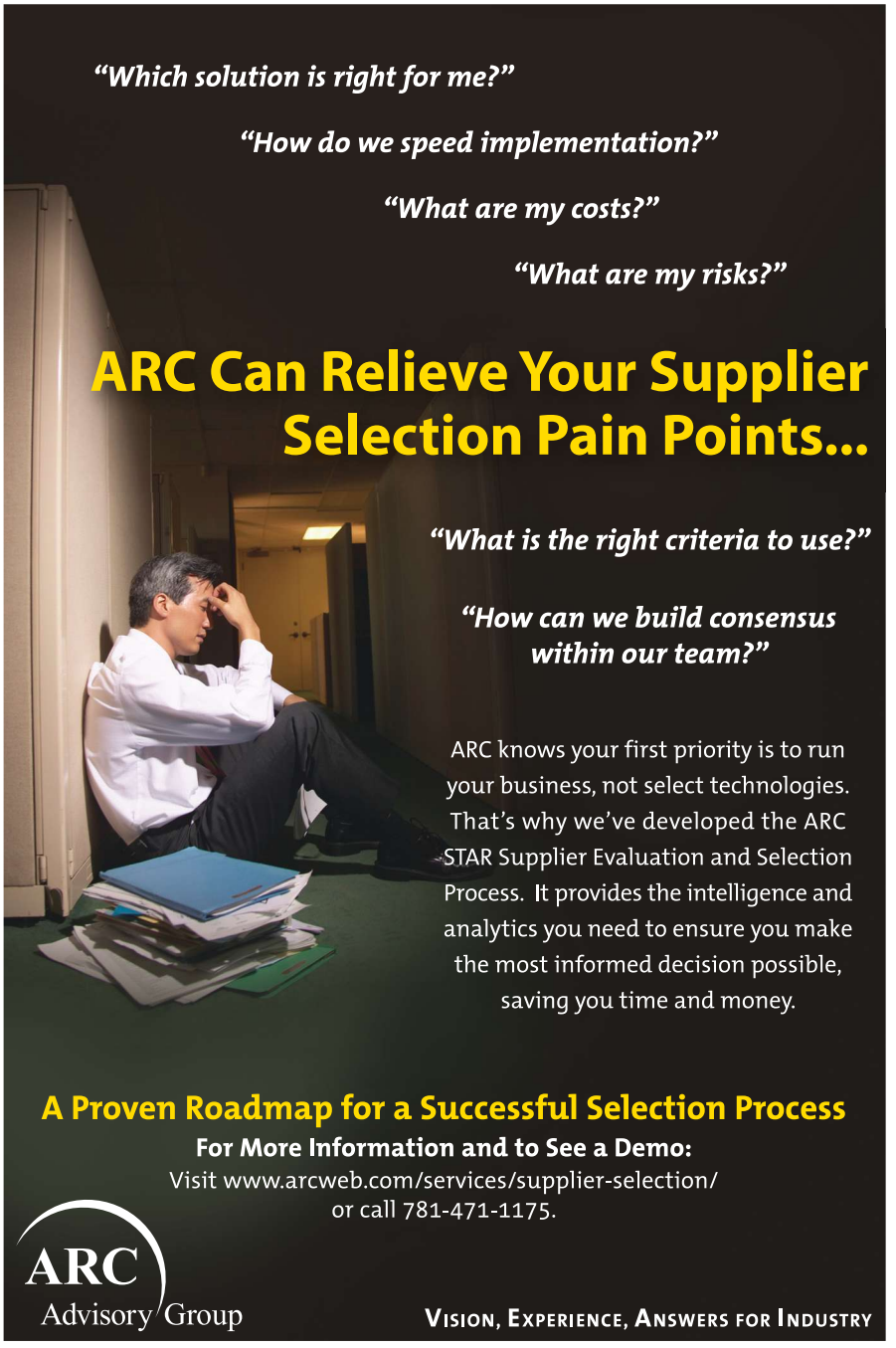

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Facilities control technician

Qorvo: This position in Richardson, Texas, provides support to the facilities operations group with process control troubleshooting and process improvement. The technician will maintain all instrumentation in the plant. This requires the ability to calibrate and install various types of instruments and understand the relationship of each system and the effect each has on the operation of the plant. Expertise in using Wonderware, OPC servers, I/O servers, and network hardware and the ability to manage the SCADA network, intelligent switches, media converters, and computer upgrades is desirable. Seven to 10 years of plant controls experience with field instrumentation is required . . . see more at ISAJobs.isa.org.

Senior application engineer

Mitsubishi Electric Automation: The senior application engineer, based in Vernon Hills, Ill., will focus on industrial/commercial engineered solutions and systems sales. The engineer will develop automation and drive-based panel solutions and system solutions for both the end user and industrial and commercial OEM markets. The engineer will work with internal and external engineering groups and interact with sales personnel in business development, regional sales, and distributor sales. The position requires 30 percent travel, machine or process control panel design and build experience, familiarity with NEC and UL standards, hands-on experience with AutoCAD . . . see more at ISAJobs.isa.org.

Engineer I, II, III

St. Johns River Water Management District: The engineer in Palatka, Fla., does water quantity and water quality modeling of surface water basins using major watershed models such as HEC-HMS, HEC-RAS, SWMM, BASINS/HSPF, and district-developed models. He or she also performs computer model calibrations and verifications of watershed simulations; maintains, updates, and modifies water quantity/quality simulation models to evaluate alternative surface water management plans; and performs hydraulic simulation of streams and rivers using hydraulic models to evaluate the risk of flooding. Occasional field work (land and water) is required . . . see more at ISAJobs.isa.org.

Control system technician

Strathcona County: The technician, working in Alberta, Canada's capital region, is responsible for providing technical support in planning and coordinating the capital and operation upgrades and modifications to the supervisory control and data acquisition system. He or she will strive for continuous improvement through incorporating and integrating emerging technologies in PLC hardware, software, and communication installations. Qualifications include a diploma in electrical or instrumentation technology or CCST with CET designation, a minimum of three years of experience related to SCADA systems, and comprehensive knowledge in programmable ladder logic within SCADA systems . . . see more at ISAJobs.isa.org.

Six key steps for an effective process change

By Chuck Boyd

ABOUT THE AUTHOR

Chuck Boyd (chuck.boyd@beamex.com), business development director for Beamex, has been involved in software development, system implementation, and business development roles for the past 29 years. Boyd was a business system analyst on software development projects for 10 years, where he was a communication conduit between the business and IT. Today, Boyd manages key accounts and market development in the southeastern U.S.

You probably use dozens of business processes every day. When processes work well, they can significantly improve efficiency, productivity, and customer satisfaction. You have also likely realized the results of inefficient processes that cause frustration, delays, and financial loss. That is why it is so important to improve processes that are not working well.

Processes can be formal or informal. Formal processes are documented and have well-established steps. Informal processes are more likely to be ones that you have created yourself, and you may not have written them down. These different kinds of processes have one thing in common: they are designed to streamline the way that you and your team work.

Keep in mind that you will need to improve most processes at some point. New goals, new technology, and changes in the business environment can all cause established processes to become inefficient or outdated. In practice, few organizations have the luxury of redesigning their processes from "a clean sheet of paper." People, equipment, and business knowledge cannot be easily scrapped.

The path to change can be a stumbling block if we assume technology is independent of the environment and organizational structure in which it is to be implemented. By some estimates, 70 percent of such projects fail to reach their intended goals. The program that seeks to become a "house of quality" can turn out to be a "house of cards" without recognition of interdependencies among technology, processes, and people. Manufacturers cannot, and will not, be experts in every aspect of software that can improve their businesses, so they must find the most suitable system and, most importantly, the best vendor as a partner to guide them through the implementation process.

Let's look at a calibration system implementation where 70 to 80 percent of the implementation is focused on process and business culture and a relatively small portion on the technology itself. The first step is to outline a system implementation process. There are a few overarching fundamentals:

- There will be a large number of interrelated tasks, carried out in a particular sequence, with decisions being made throughout.
- A proven implementation model will yield quicker results than developing a model from scratch.
- The project must be overseen by a dedicated project manager with committed involvement by key resources.
- Poor planning and inadequate resourcing are the cause of problems such as scope creep and budget and schedule overruns.

You can successfully implement a new calibration process by following a well-defined, step-by-step implementation process:

1. Initiation: Define the entire framework for the project. Define the scope of the new system and launch date, appoint a project team, prepare a project plan, define change management procedures and testing/approval procedures, and establish the roles and responsibilities of both the manufacturer and vendor.
2. Process blueprinting: Map the existing calibration process and reengineer it for the target system. Explore each phase in detail, as some processes may have substeps that you are not aware of. It is critical to consult people who use the system regularly, as well as cross-functional experts such as information technology, compliance, and quality. At this point, the project can still be canceled.
3. Specification: Using the blueprint as the basis, define the user requirements, functional design, and test specifications. In this step, it is imperative that the manufacturer and vendor clearly understand the technical specifications and functionalities they are agreeing to.
4. Execution: Execute the plans prepared during the specification phase. This phase is where change management procedures may be used.
5. Deployment: Deliver all software, equipment, and documentation as well as the customer's acceptance of delivery. Train target groups, and complete and make standard operating procedures, work instructions, and process descriptions available. The system is transferred from the project team to the operational team.
6. Operation: Use the new system in production and realize the improved process. Support agreements become effective.

Efficient business processes could not be more important in the current climate of competitive global markets, subdued demand, and competitive pricing. Where processes heavily involve interaction with the workforce, such as the calibration implementation example, efficiency becomes even more critical. As a final note, there is no end point; optimizing business process efficiency will always be an ongoing process. ■

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